RECURSIVE ORGANIZATION IMPROVEMENT: A MODELING SPECIFICATION FOR HUMAN–AGENT ORGANIZATIONS

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ABSTRACT

Stronger AI agents do not automatically produce better organizations: teams must also learn which work arrangements to retain and when to reconsider them, while those arrangements determine what evidence the team can learn from. We propose a modeling specification for recursive organization improvement and evaluate it through an executable checker, a public-record mapping, and two simulation studies. The specification connects actor-visible histories, organizational memory, decision rights, and the evidence used to retain changes. Study 1 crosses six decision rules, three memory conditions, and three task environments under fixed resource ceilings. In a stationary environment, cumulative evidence raises balanced evaluation's net value per task from 0.452 to 0.480, while selection regret falls from 0.029 to 0.002. After a workflow-ranking reversal, retained evidence delays adaptation. Acquisition-matched controls leave small, uncertain differences between fixed and replaceable evaluators across three reversal timings. Study 2 models human–agent review with shared errors, copying, and decision rights. It identifies a two-actor case in which public records conceal whether agreement comes from shared errors or copying, and shows how dependence changes the preferred placement of human judgment under a fixed budget. Paid observation can lower decision error while costing more than the deployment gain. Assessments of organizational improvement should track the evidence acquired and retained, using acquisition-matched comparisons to isolate the contribution of evaluator revision.

Keywords human–AI interaction · multi-agent organizations · organizational learning · workflow adaptation · recursive improvement · evaluation

1 Introduction

Consider a software team that uses AI agents to draft pull requests (PRs). Faster drafting can lengthen the review queue. The team may respond by routing routine changes to automated checks and reserving human review for difficult cases. To evaluate this change, it must decide which PRs to audit, how to combine new observations with earlier evidence, and who may revise the routing or audit rule. An apparent gain can reflect a better workflow, a different sample of reviewed cases, or more effective reuse of evidence. How can the team distinguish these explanations and retain changes that improve delivery?

We propose a modeling specification for this problem and evaluate it with an executable checker, a mapping of public PR records, and two simulation studies. The running PR example connects the specification's fields to concrete organizational decisions; the studies isolate evidence acquisition and retention, then the review arrangements that generate observable judgments.

Task-level productivity studies and studies of human–AI collaboration address different parts of this problem. Experiments on professional writing and evidence from customer support document productivity effects in particular settings [11, 36]; a meta-analysis finds substantial variation in the performance of human–AI combinations relative to their

constituent members [44]. Organization design supplies the intervening objects: the division and allocation of work, dependencies, and integration of effort [32, 39]. Complementary organizational investments can also affect the returns to a new technology [10, 34]. Consequently, comparing agent models while holding the organization fixed answers a different question from comparing organizations that can revise their workflows.

Workflow revision creates a measurement problem of its own. A new review policy changes both which cases receive attention and which failures become visible. If the same selected cases determine whether that policy is retained, organizational learning can favor a locally successful but globally harmful arrangement. An improvement procedure therefore needs an explicit account of the evidence it can acquire, the changes it can authorize, and the costs it incurs. The procedure may itself become a target of revision. Study 2 examines how review visibility and decision rights shape the evidence available for judging an arrangement.

We call this problem *recursive organization improvement* (ROI; distinct from return on investment): evaluating and retaining changes to organizational arrangements, including changes to the procedures through which those arrangements are improved. The research object is the human–agent organization. Agent capability, human judgment, coordination, evidence access, and authority can change at different rates and under different constraints.

This paper contributes a modeling specification that links organizational state, actor-visible histories, and the records used to accept a change. A revision names both the arrangement being changed and the procedure used to evaluate it. This makes it possible to reconstruct the evidence available at a decision, check editing rights and resource limits, and compare the outcomes of retained changes.

Study 1, *evidence acquisition and retention*, holds label quality fixed and crosses reset, cumulative, and finite-window memory with fixed evaluation programs, one-time discovery, and repeated program assessment. Retention removes much of repeated assessment's disadvantage in a stable environment but delays adaptation after a workflow-ranking reversal. The acquisition-matched control is central to interpreting the apparent gain from evaluator revision: a fixed program receives the same trials and reusable labels, leaving a small and uncertain difference attributable to program replacement.

Study 2, *review arrangements and evidence generation*, derives organizational outcomes and visible judgments from individual errors, shared error sources, upstream exposure, and decision rights. Its contribution is to distinguish mechanisms that, in a two-actor case, can look identical in public records but respond differently to blinding, separate gold-labeled measurement from human override authority, and show how error dependence changes the preferred placement of humans under a fixed budget. Paid observation policies test whether identifying a better arrangement also recovers the cost of learning about it.

Study 1 follows evidence through discovery, retention, and reassessment. Study 2 examines how that evidence is generated: blind review preserves private judgments, whereas exposed review can replace them with copied agreement. A team choosing a review arrangement therefore also chooses which judgments its improvement procedure can observe. Study 2 compares fixed observation policies for learning about these arrangements.

2 Conceptual foundations and modeling requirements

2.1 Organizational learning and explicit organizations

Organizational learning connects experience to retained routines [29, 33]. The distinction between the abstract understanding of a routine and its situated performance helps explain how routines can generate both stability and change [16]. Deliberate learning adds articulation and codification of experience [47], while double-loop learning addresses governing assumptions and values [4]. Engelbart's improvement-of-improvement distinction supplies a close conceptual antecedent for examining the procedures that generate organizational change [15]. ROI translates this question into versions, events, change targets, and outcome comparisons.

Human–agent organizations also require an account of interdependence. Mixed-initiative interaction and levels of automation distinguish who initiates or performs a function [22, 37]; Coactive Design places the support for interdependent activity at the center of system design [25]. Appropriate reliance concerns when a person should use automated advice [28]. These choices alter information exposure, decision rights, and resource use, so they belong in the organizational model rather than being inferred from an actor's capability score.

Existing formalisms already supply much of the representational machinery. MOISE+ models structural, functional, and deontic organization; OperA represents organizational interaction and agent autonomy [14, 24]. Process mining uses event data for process discovery, conformance checking, and enhancement [45]. The proposed specification connects these objects to the evaluation and retention of changes through the relations in table 1. A sufficiently annotated event log or state-transition model can encode the same relations. The contribution is their explicit organization into a comparison contract, whose completeness and execution can be checked.

Table 1: Connections to established approaches. The right column identifies the records used in an ROI comparison.

Approach	Established modeling object	Required connection for a change comparison
MOISE+; OperA	Roles, missions, norms, interactions	Target-specific authority, prior version, evidence and comparator for a revision.
Process mining	Cases, activities, event order, conformance	Organization/procedure versions, actual observation access, and sampling coverage.
Organizational learning	Experience, routines, retention	Link a retained routine to the experiment and criterion used to accept it.
Agent/workflow search	Executable candidates and evaluations	Human resources, rights, generated-output costs, and revision of the evaluator.
Generic event log	Arbitrary recorded attributes	The same relations, when recorded; missing fields remain unknown.

Automated agent-system design and workflow optimization provide candidate-generation mechanisms [23, 46]; linguistic feedback can support behavioral revision [41]. In ROI, these mechanisms are embedded in an organization with human roles, restricted evidence, and retention decisions. The central comparison concerns organizational outcomes under specified changes, whether the candidate was written by a person, generated by an agent, or selected from a finite catalog.

2.2 From development practices to measurable arrangements

Waterfall-like staging, agile development, and Extreme Programming suggest different settings for dependencies, batch size, feedback delay, test placement, and ownership. Royce's original account includes iteration and the risks of late system testing [40]; agile principles emphasize frequent delivery and reflection [8]; XP provides more concrete testing, pairing, and integration practices [2]. These labels identify families of configurations rather than an ordering of organizational quality. A team can retain release gates while shortening internal feedback cycles, or introduce agents into an unchanged approval pipeline.

For example, let drafting, review, and integration capacities be 4, 5, and 9 accepted-task equivalents per day. A deterministic serial capacity bound is their minimum. Doubling drafting alone raises the bound from 4 to 5; raising review capacity to 8 at the same time raises it to 8. The factorial interaction is $8 - 5 - 4 + 4 = 3$. This standard bottleneck calculation [3, 30] isolates why technical and workflow changes can be complementary. Rework, variable quality, and demand must be modeled separately in an empirical application. Our specification records the fields needed to distinguish such joint changes from an increase in task-level speed.

3 A modeling specification for recursive organization improvement

3.1 State, improvement procedure, and change boundary

An instance declares a task population, environment process, outcome criterion, and assessment horizon. The environment generates tasks and outcomes; actor policies receive only the observations assigned to them. At event index t, define

$$\mathcal{O}_t = (X_t, D_t, L_t, \Phi_t, \Pi_t, M_t), \qquad U_t = (g_t, a_t, e_t, s_t, r_t). \tag{1}$$

The organization state $\mathcal{O}$ describes who does the work and what they can know or change. The procedure U describes how the team proposes, tests, selects, and retains changes. Table 2 maps their components to the PR example. Actors also have task-specific capability profiles; approval rights are recorded separately.

Definition 1 (Declared boundary)**.** *A boundary $\mathcal{B}$ partitions fields into operational arrangements, improvement-procedure fields, and externally fixed fields for a stated comparison. It identifies authorized editors, the evaluation horizon, and the outcome criterion.*

An execution event changes work artifacts. An operational adaptation changes an arrangement under the declared improvement procedure. A procedure revision changes a field of U and evaluates its consequences for subsequent organizational changes or outcomes. For instance, assigning a PR to a reviewer is execution; replacing its routing rule is operational adaptation; replacing the sampling program used to evaluate routing changes is procedure revision. A fixed outer selector may govern all three. The boundary $\mathcal{B}$ identifies which changes count as procedure revisions.

This boundary also resolves overlapping implementations. Φ applies the current visibility rules, while a requests evidence for a change evaluation. If one patch affects both, the implementation records the coupled fields in the state

Table 2: Core notation, illustrated by PR review routing.

Symbol	Meaning	PR example
X	Actors, roles, tasks, versioned artifacts	Reviewer, agent, PR, test report.
D	Task and resource dependencies	Approval depends on tests and reviewer availability.
L	Decision rights	Maintainer may change routing; review board may change auditing.
Φ	Observation availability	Which test results or private assessments each actor can see.
Π	Interaction and allocation protocols	Assignment order, review sequence, and workload allocation.
M	Retained evidence and versions	Audit labels, trial scores, and the adopted routing rule.
$U = (g, a, e, s, r)$	Generate, acquire, evaluate, select, retain	Propose a rule; audit PRs; score and select it; monitor continued use.
$\mathcal{B}$	Declared change boundary	Routing and auditing may change; outcome criterion stays fixed.
$\mathcal{H}_{i,t}$	Evidence visible to actor i before event t	Reports available before approval.
$p, \mathcal{T}, H$	Change contract, event trace, evaluation horizon	Recorded routing change and its follow-up period.

and cost ledger. If the outcome criterion changes, its version changes too; comparisons under the old and new criteria are reported separately.

3.2 Events and actor-visible histories

For the PR team, an event may be a reviewer assignment, a private assessment, or a revealed test report. An event z records an identifier, logical order, actor, action, input/output artifact versions, recipients, resource use, and organization version. Actions include assignment, commitment, reveal, critique, test, escalation, approval, and patch application. Logical order represents dependency; physical timestamps can additionally represent concurrency and elapsed time.

Let $\mathcal{H}_{i,t}$ contain actor i's permitted initial information and artifacts revealed to that actor before event t. A policy satisfies

$$b_{i,t} \sim \pi_i(\cdot \mid \mathcal{H}_{i,t}, \mathcal{O}^{\text{vis}}_{i,t}), \tag{2}$$

where $\mathcal{O}^{\text{vis}}_{i,t}$ is the actor-visible projection of the organization state. Thus a simulator can score a decision using a hidden true error rate while the decision maker receives sampled labels only. A seed supports reproducibility without granting actors access to each other's random choices.

The ordering of commitment and reveal events represents human-first, agent-first, and parallel assessment. Parallel commitments preserve the opportunity for private judgments, while shared training data, common tools, or correlated task difficulty can still induce dependent errors. Statistical dependence is a joint property requiring measurements or explicit assumptions; it cannot be inferred from separate model names. A public review timestamp records publication; establishing what another actor read requires exposure records.

3.3 Change contracts and retention

A proposed routing change carries a contract: a record linking the change to its evidence, comparator, cost, and follow-up rule. Formally,

$$p = (\tau, v, \delta, \mathcal{E}, \mathcal{C}, c_p, H, \mathcal{R}_p). \tag{3}$$

τ identifies the target and proposer; v is the required input version; δ is the transformation; $\mathcal{E}$ identifies available evidence and its provenance; $\mathcal{C}$ states the comparator, population, and criterion; c_p specifies resource accounting; H is the horizon; and $\mathcal{R}_p$ specifies retention, monitoring, and rollback conditions. Contract fields may refer to separately stored artifacts. Their relationships, summarized in table 3, determine whether a comparison can be reconstructed.

Admission checks authorization, versions, available evidence, prerequisites, and resources. Efficacy is a subsequent outcome comparison. These predicates have different witnesses: permission records can justify admitting a trial, while a trial's results can justify retaining its policy. Rejected and unsuccessful revisions remain in memory because they consume resources and influence future search.

For external criterion J and specified comparator 0, the effect of a patch is

$$\Delta_H(p) = \mathbb{E}[J(\mathcal{T}^p_{t:t+H}) - J(\mathcal{T}^0_{t:t+H})] - C_H(p). \tag{4}$$

Table 3: Specification obligations and observable failure conditions. Unknown information is distinguished from a violated condition.

Relation	Required condition	Failure or missing record
Actor–evidence	Each decision input was available to its actor beforehand.	Hidden/future input; unknown exposure history.
Patch–state	The target version exists and matches the expected version.	Stale patch; unversioned configuration.
Actor–target	The actor holds the applicable editing right at that version.	Unauthorized change; unknown historical permission.
Evidence–population	Sampling scope and label source support the stated population.	Uncovered stratum; unknown inclusion mechanism.
Outcome–criterion	Both arms use the declared criterion and horizon.	Changed success definition; censored follow-up.
Cost–resource	Search, outputs, labels, and transitions enter one ledger.	Omitted counterfactual output; duplicate charge.
Decision–memory	Accepted and rejected proposals retain evidence and versions.	Retained policy without its validation domain.

C_H contains costs not already included in J. Randomized comparisons, controlled simulation, or explicit causal assumptions are needed to estimate the two potential outcomes. Outcomes without defensible common units remain a vector with declared constraints. The simulation criteria below include all modeled costs in net value, giving $C_H = 0$ in equation (4).

Figure 1 connects execution to this revision loop. A software team might first change review routing, then discover that its evaluations cover only human-reviewed PRs. A second patch can alter evidence acquisition to include routine changes. That second patch is evaluated through later routing decisions and their outcomes.

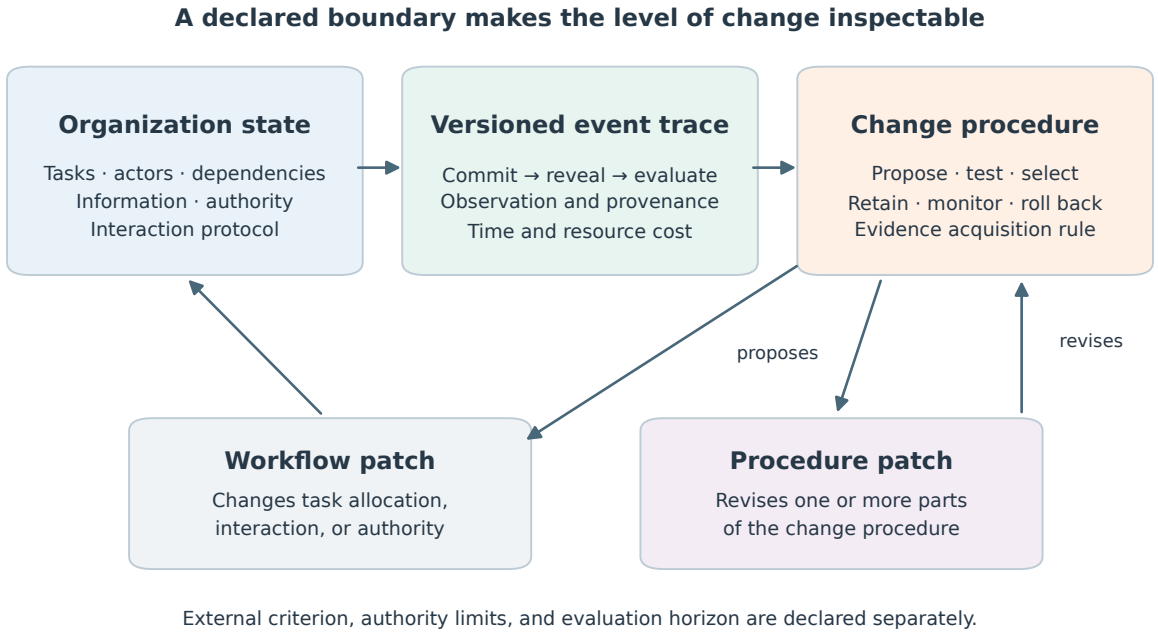


Figure 1: The modeling specification connects organization state, event histories, and improvement procedures. A revision names its target, evidence, authority, and external evaluation criterion.

3.4 Organizational memory as a testable design choice

The memory field M separates three retained objects: workflow evidence, evaluation-program evidence, and adopted arrangements. Keeping a program name preserves the adopted decision; accumulating knowledge requires retaining its

supporting observations. Keeping observations indefinitely can preserve knowledge about a population that no longer exists. An instance therefore records evidence provenance, applicability, and expiry alongside the retained policy.

The retention component $U.r$ specifies when evidence or arrangements are reconsidered. A fixed window is one such rule; changing the rule is a procedure revision at the declared boundary. In Study 1, memory rules are experimentally assigned and held fixed within a run. Evaluation programs may then change inside each memory condition. This factorial construction separates a design intervention on retention from the execution of a program-selection procedure.

Evidence generated during organizational search can serve more than one decision. A rejected program trial may still produce valid labels about a workflow. The trace links those labels to their source and reveals them to the permitted downstream evaluator. Comparing a policy that reuses those labels with one that discards them tests an information-transfer rule. Comparing two policies with the same trials and label reuse, while allowing only one to replace its program, tests a more specific contribution of program revision.

3.5 What the implementation checks

The reference implementation provides a finite checker for unique event/artifact identifiers, visible inputs, version matching, protected criteria, target-specific permissions, and patch budgets. Constructed fixtures accept a valid trace and patch and reject a hidden input and five invalid patch variants. A separate target test allows a release owner to change routing but rejects the same actor's audit-rule change; granting the review board that right admits it. This checks the operational consequences of the declared boundary.

Study 1 instantiates a narrower executable subset: retained label statistics, retained trial scores, program versions, authorized patch application, label budgets, and a modeled cost ledger. It exports trajectory outcomes, program proportions, conditional block outcomes, and example version transitions. Full implementations would additionally need to enforce all relationships in table 3, including criterion and retention contracts. The checker tests these admission conditions; the simulations compare the outcomes of accepted changes.

Study 2 instantiates parts of Actor–evidence through purchased private and exposed judgments, Cost–resource through acquisition, production, and switching charges, and Decision–memory through retained observation counts and recorded arrangement choices. Study 2 selects arrangements directly from a fixed catalog, outside the versioned-patch interface of the reference checker.

Mapping public repository records supplies a complementary feasibility check. Three merged Flask PRs yield four commit artifacts and 56 mapped events: eight reviews, 42 head-commit check runs, and six opening/merge records. Artifact identifiers and recorded actions are recoverable. Historical authority, actual message exposure, effort, organization versions, and AI involvement remain unknown. Appendix A specifies the snapshot and mapping. Such a trace can expose instrumentation gaps before a team attempts a workflow-effect estimate.

4 Study 1: Acquiring, retaining, and reconsidering evidence

Scope of the simulations. Both studies examine evaluation and review arrangements in human–agent organizations using synthetic task, actor, and cost parameters. The core Study 1 design preceded execution; its acquisition-matched and change-timing controls were specified after inspection of the core results. Study 2 was designed after Study 1 and is exploratory. Task decomposition, capacity, queues, and specialist–generalist trade-offs are represented by the specification but lie outside these simulations.

4.1 Research questions and organizational instance

Study 1 follows a change from its discovery to its later use. It asks whether evidence retention alters the value of evaluator revision, whether a one-time discovery can substitute for repeated search, and which part of reassessment produces a measured gain. These questions instantiate $U.a$ (evidence acquisition), M (retained evidence), and $U.r$ (continued use or reconsideration) in the specification.

The relations in table 3 constrain specific design decisions. Decision–memory requires separate records for workflow labels, program scores, and the adopted program; we therefore intervene on these memories separately. Cost–resource requires each queried label to enter one ledger, even when both program trials and workflow selection reuse it. Actor–evidence requires current program trials to share the same pre-review information and prevents access to hidden error probabilities. Outcome–criterion keeps the deployment horizon and net-value rule common across arms. Study 1 uses these obligations to distinguish a change in the evaluator from a change in its evidence.

A proposer supplies three workflow templates, an evaluator acquires labels, and a review board may replace the evaluation program. In Study 1, the roles denote information access and decision rights and can be assigned to humans or agents; Study 2 models their individual judgments (Section 5). The environment has two equally frequent

Table 4: Decision rules and the comparisons they support. All arms use the same memory condition and block-level resource ceiling.

Arm	Program decision	Role in the study
Biased / Balanced / Neyman	Program specified before execution	Fixed designs with different coverage/allocation rules.
Successive rejection	Fixed elimination rule across templates	A between-template elimination comparator.
Discover once	Compare programs once; retain the winner	Discovery followed by reuse and budget reallocation.
Repeated discovery	Reassess every eight rounds	Incremental value of continued program assessment.
Trial-matched Balanced	Run the same trials; keep Balanced	Exploratory control for evidence acquisition and timing.

observable task strata. Error probabilities are $(.20, .20)$ for the standard template, $(.16, .16)$ for broad, and initially $(.08, .46)$ for specialized. Thus broad is initially best, while specialized helps one stratum and harms the population. In *stationary harm* these probabilities remain fixed. In *workflow reversal*, specialized becomes $(.08, .08)$ before round 25 and becomes the best template. *Uniform gain* uses that improving specialized template throughout.

Each of 48 rounds deploys one template on 4096 production tasks. Only acquired evaluation labels enter decisions; the external scorer uses exact expected production outcomes. This separates uncertainty in organizational judgment from production sampling noise. One value unit is the gross value assigned to one completed production task before errors and costs. Net value per task subtracts error penalties and modeled expenses in these normalized units. An error costs 3 units, generation costs .02 per production or evaluation output, a label costs .20, and changing evaluation programs costs .50. Storage and computation costs are outside this ledger.

4.2 Memory and decision rules

The population-weighted posterior error estimate of template k is

$$\widehat{r}_{k,t} = \frac{1}{2}\sum_{s=1}^{2}\frac{1+E_{ks,t}}{2+N_{ks,t}}, \tag{5}$$

where E and N count observed errors and labels available to the decision. All conditions begin with $\mathrm{Beta}(1,1)$. *Reset* uses current-round evidence only. *Cumulative* retains all earlier evidence. *Window 8* retains evidence from the preceding eight rounds plus current observations. The same retention rule applies to past evaluation-program scores; a program's identity remains retained until another is selected. This separates remembering a decision from remembering the evidence supporting it.

Three evaluation programs form a fixed catalog. Biased allocates 95% of a template's allowance to the first stratum. Balanced uses equal counts. Neyman uses a two-label-per-cell pilot and posterior variance estimates to allocate the remainder across strata [35]. Each program chooses the template minimizing equation (5). Counts use population weights regardless of sampling proportions.

Workflow selection is a fixed-budget identification problem: evaluation outputs are acquired to choose a template for subsequent deployment. Successive Rejects addresses this objective by progressively eliminating candidates [5]; exploration sampling explicitly targets policy choice rather than immediate experimental outcomes [26]. We use a between-template elimination comparator based on successive rejection, whereas Neyman allocates within each template. It observes paired stratum labels, eliminates the worst estimated candidate after a first phase, and compares the remaining two after a second phase. Historical evidence enters the same posterior estimates under the assigned memory rule. Allocation formulas and tie rules appear in Appendix C.

4.3 Discovery, reuse, and fixed resource ceilings

Table 4 distinguishes knowing an evaluator, discovering it once, and periodically reconsidering it. A label unit is one observed binary error outcome for one evaluated output. Every arm receives a ceiling of 2736 such observations per eight-round block. A fixed arm can allocate 342 units per round, or 114 per template for the three catalog programs. Discover once pays for program comparisons only before round 1. Repeated discovery compares programs before rounds 1, 9, 17, 25, 33, and 41. Both begin with Biased and select among the same catalog.

In the base design, program trials receive at most 20% of the block ceiling, with one trial per catalog program. Each trial screens the three workflows under the same pre-review memory snapshot, selects one, and validates it on fresh

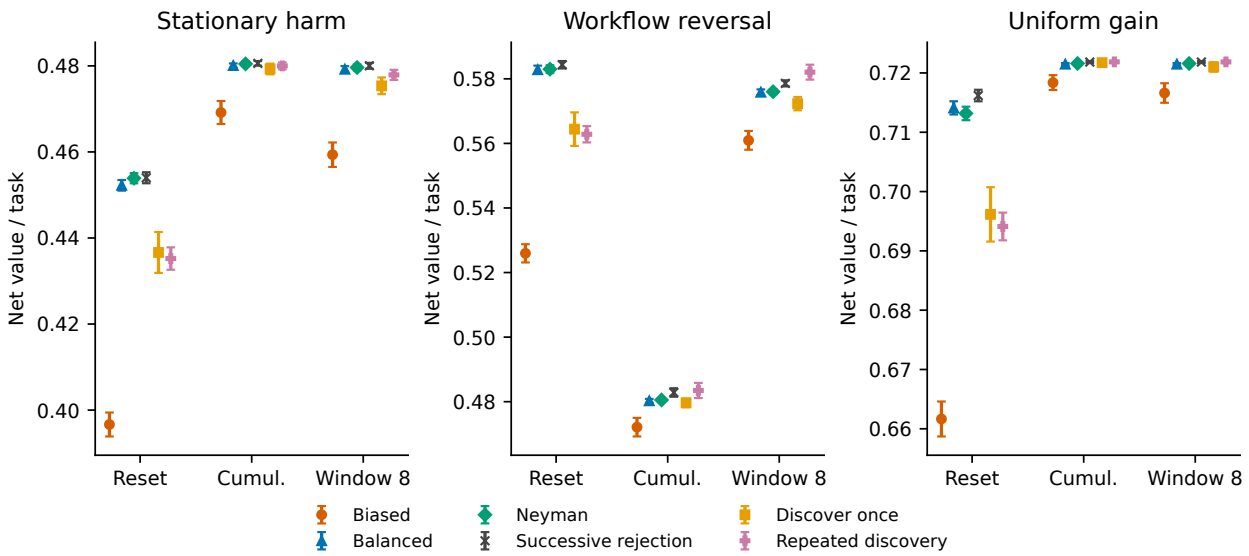


Figure 2: All core comparisons: 128 replicate trajectories per cell, means and 95% Monte Carlo intervals. Each panel uses its own value scale. "Cumul." retains all acquired evidence; Window 8 retains the preceding eight rounds plus current evidence. The results concern the jointly specified memory, discovery, and acquisition rules.

balanced labels. Its score is validation value less screening expense per production task. The highest mean retained trial score determines the program. This fixed outer selector revises the retained program within the catalog.

All acquired trial labels, including evidence from rejected programs, can enter workflow memory. Decisions about program scores precede pooling the current trials; the subsequent operational screen receives the pooled evidence. This makes failed searches potential sources of reusable information. Actual trial expenditure is deducted from the current block, and the remaining allowance is spread across its operational screens. Discover once receives the full operational allowance from block 2 onward. No arm borrows from a future block; unused units due to integer allocation are not charged.

Net value includes production generation, all acquired labels, all evaluated outputs, and program changes. Selection regret measures the gross value gap to the best available template, separately from those costs. Harmful adoption counts choices worse than standard; after reversal, this measure becomes zero for all templates, so regret and adaptation trajectories are needed to distinguish them. The primary outcome averages net value over all 48 rounds, with final-eight-round performance as a separate endpoint.

The core matrix crosses three environments, three memory rules, and six arms, using 128 independent replicate streams per cell. A separate 96-replicate grid crosses program-budget shares $.2, .4, .6$ with trial counts $1, 3, 6$, keeping the total ceiling fixed. It also removes program-score memory or trial-label pooling separately. The acquisition-matched controls use independent streams and include reversals before rounds 21 and 29 to assess attribution and timing. Appendix C specifies all conditions.

4.4 Accumulation changes the apparent cost of improvement

Figure 2 reports every core arm and memory condition. Under stationary harm, cumulative evidence raises Balanced's mean net value from $.452$ to $.480$, a paired gain of 0.0278 (95% Monte Carlo interval 0.0266 to 0.0291). Its final-eight-round value reaches $.482$, the value of consistently selecting broad under its actual evaluation expenditure. Balanced's mean selection regret falls from $.0294$ to $.00156$ per task, while its evaluation expense remains $.0184$. The net-value gain therefore equals the reduction in selection regret.

Repeated discovery is 0.0170 below Balanced with reset evidence (interval -0.0196 to -0.0144). With cumulative evidence, both means round to $.480$ and the paired interval spans zero. Discover once reaches $.479$, while successive rejection reaches $.481$. Reusing acquired evidence thus removes most of the reset disadvantage, and fixed selection rules remain competitive with program revision. Under Uniform gain, specialized is best from the outset, so retention improves selection throughout.

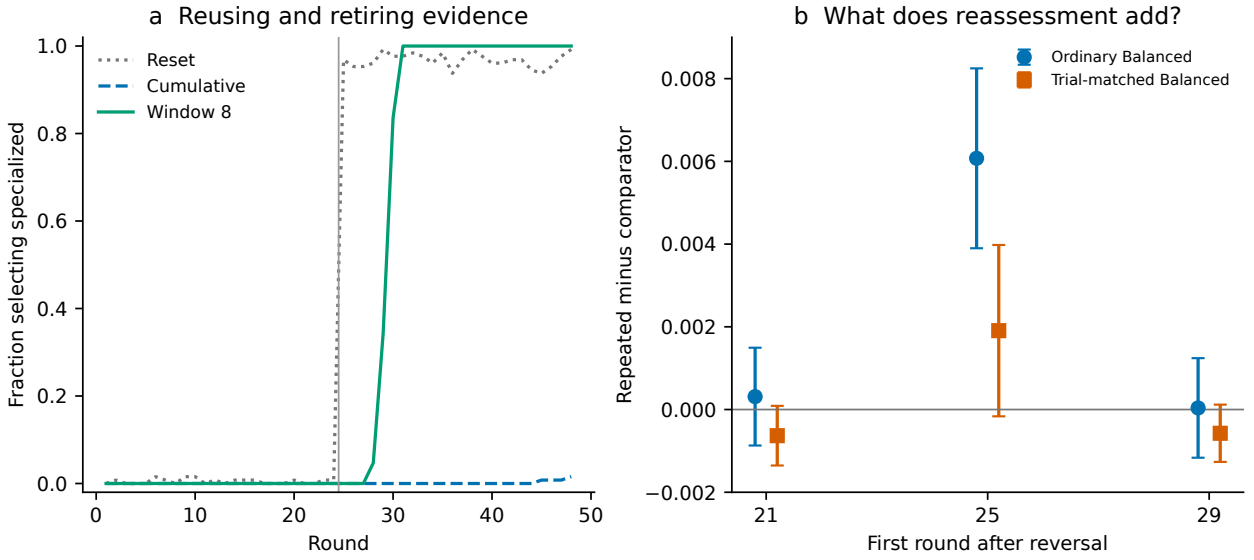


Figure 3: Retention and attribution. (a) Fixed Balanced under the core reversal; the vertical line marks the change before round 25. (b) Exploratory controls with Window 8 and reversals before rounds 21, 25, or 29. Points show repeated-discovery minus comparator differences with paired 95% Monte Carlo intervals on 128 fresh replicate streams. Matching the trials, label reuse, and timing changes the attribution of the apparent reassessment gain.

The comparison to fixed programs also depends on prior knowledge. An equal-probability initial draw from the three catalog programs gives net values of .434 with reset evidence and .477 with cumulative evidence, compared with .435 and .480 for repeated discovery. Table 6 reports this mixture in every core condition. Balanced provides a prespecified representative-sampling reference for distinguishing discovery from the revision of an already effective evaluation rule.

4.5 Matching acquisition isolates program replacement

Under workflow reversal, indefinite accumulation slows the replacement of broad by the newly superior specialized template. Balanced's full-horizon values are .480 with cumulative memory, .576 with Window 8, and .583 with reset evidence. The finite window improves considerably on unlimited accumulation, yet its transition delay makes it worse than reset over this particular horizon. In the final eight rounds it reaches .722, compared with .712 for reset and .483 for cumulative memory. Figure 3a displays the corresponding deployment choices.

With Window 8 in the core reversal, repeated discovery exceeds discover-once by 0.0098 (interval 0.0071 to 0.0124) and ordinary Balanced by 0.0062. Program trials also concentrate acquisition near review times and supply labels that workflow selection can reuse. The trial-matched control gives Balanced these same trials, pooled labels, and acquisition timing while keeping its program fixed. On independent streams with reversal before round 25, the repeated-discovery advantage falls from 0.0061 over ordinary Balanced (interval 0.0039 to 0.0082) to 0.0019 over trial-matched Balanced (interval −0.0002 to 0.0040). At reversals before rounds 21 and 29, the remaining mean differences are slightly negative and both intervals span zero. Acquisition matching therefore leaves a small, uncertain contribution from program replacement across these three timings.

4.6 What the allocation and retention diagnostics explain

The full nine-cell budget-share/trial-count grid is reported in table 7. Program trials can improve subsequent workflow selection by adding reusable labels to workflow memory. Removing trial-label pooling lowers repeated discovery from .480 to .478 under stationary cumulative memory, and from .582 to .574 under reversal with Window 8 on the independent sensitivity streams.

Table 8 relates the retained program to subsequent-block outcomes. At the last stationary review with cumulative memory, Biased, Balanced, and Neyman are retained in 38.3%, 31.2%, and 30.5% of runs. The 40 runs retaining Balanced have subsequent-block net value .482 and zero harmful adoption. Accurate workflow selection can therefore coexist with different retained programs when their decisions share a well-informed memory. Supplementary Table S2 summarizes the final program-selection proportions across environments and memory conditions.

5 Study 2: How review arrangements generate evidence

5.1 Research questions and connection to Study 1

Study 1 prescribes workflow error rates and label quality independently of the arrangement that acquires evidence. Study 2 makes the expressed judgments and final outcomes depend on a human–agent review arrangement. The same private signals may remain available under blind review or be replaced by copied agreement after exposure. Proposers and reviewers instantiate X; exposure and order change Φ and Π; arbitration changes L; and observation retention instantiates M.

We vary private error dependence and copying at fixed individual accuracy, compare review arrangements under a common budget, and test observation policies for learning which arrangement to deploy.

5.2 Private errors, copying, and bounded review arrangements

A binary task has a proposer, agent reviewers, and, in some arrangements, a human arbiter. Humans have private error probability q_H; conditional on task truth, their errors are independent of one another and of the agent group. Agents have private error probability q_A. With probability ρ_A, all agents on a task, including an agent proposer, share one error draw; otherwise their errors are independent. This construction preserves individual accuracy while varying private error dependence.

Motivated by observed sycophancy in multi-agent debate [38], we isolate copying (echo) without additional task information. Here η_s is the probability that an agent reviewer copies an upstream decision from source $s \in \{H, A\}$; otherwise it retains its private judgment. The subscript denotes the upstream actor, so η_H describes an agent copying a human. Blind protocols preserve private commitments for aggregation or escalation while retaining any intrinsic error dependence. We compare exposed and blind voting, exposed and blind disagreement-triggered human arbitration, random arbitration of unanimous cases, serial review, and single-actor baselines. Exact enumeration identifies a minimum-error arrangement within the declared candidate catalog and budget; expected cost breaks error ties. The error-first grid isolates decision quality within a hard resource ceiling. The learning comparison instead selects on deployment loss $3e + .02c$, where e is error probability and c is expected production call cost; its net-value score additionally deducts observation and switching costs (equation (12)). Appendix E specifies the joint error law, protocols, costs, parameter grid, and ties.

5.3 Visible agreement and the placement of human judgment

For an agent proposer and three agent reviewers, private accuracy of 90% and $\rho_A = .1$, increasing η_A from 0 to .8 raises unanimous agreement with the proposer from 69.1% to 91.2%, while majority-decision error rises from 3.52% to 9.64%. Conditional on an incorrect proposal, unanimous approval rises from 10.1% to 59.6%. Blind voting retains an error rate of 3.52% under the same private-signal distribution. These curves illustrate the consequences of the copying rule. The identification problem appears when distinct mechanisms generate the same public record. With one agent proposer and one equally capable agent reviewer, $(\rho_A, \eta_A) = (.8, 0)$ and $(0, .8)$ produce the same joint distribution of public votes and truth but respond differently to blinding (Appendix E).

Figure 4 separates copying from pre-existing dependence. With $q_H = .2$, $q_A = .1$, a human-to-agent call-cost ratio of four, blind overhead of .05 per reviewer, and a hard task budget of three human-call units, the minimum-error choice at $\eta_H = \eta_A = .5$ changes from an agent proposer with six blind agent reviewers at $\rho_A = 0$ to a human proposer, one blind agent reviewer, and an independent human arbiter on disagreement at $\rho_A = 1$. The preferred arrangement changes where human judgment enters the decision. In the disagreement gate, a human makes the final decision on disputed cases. Learning about the gate uses a separate gold-labeled audit. The gate changes decision authority; the audit measures performance.

5.4 Paid observation, retention, and net value

We compare four fixed observation policies for selecting arrangements with unknown actor parameters. Across 48 rounds, they choose among 58 budget-feasible arrangements under reset, cumulative, or Window 8 memory. They acquire final outcomes only, paired private and post-exposure judgments, or observations from randomized blind and exposed tasks. An *Omit echo* ablation receives the same paired records but sets copying probabilities to zero in its predictions. Six environments, including a copying shift before round 25 and source-dependent copying, use 128 replicate trajectories per condition. All policies share a ceiling on acquisition cost, but purchase different numbers of labels. The cost schedule is separate from Study 1's label-count budget. Observation policies are assigned treatments and do not change during learning.

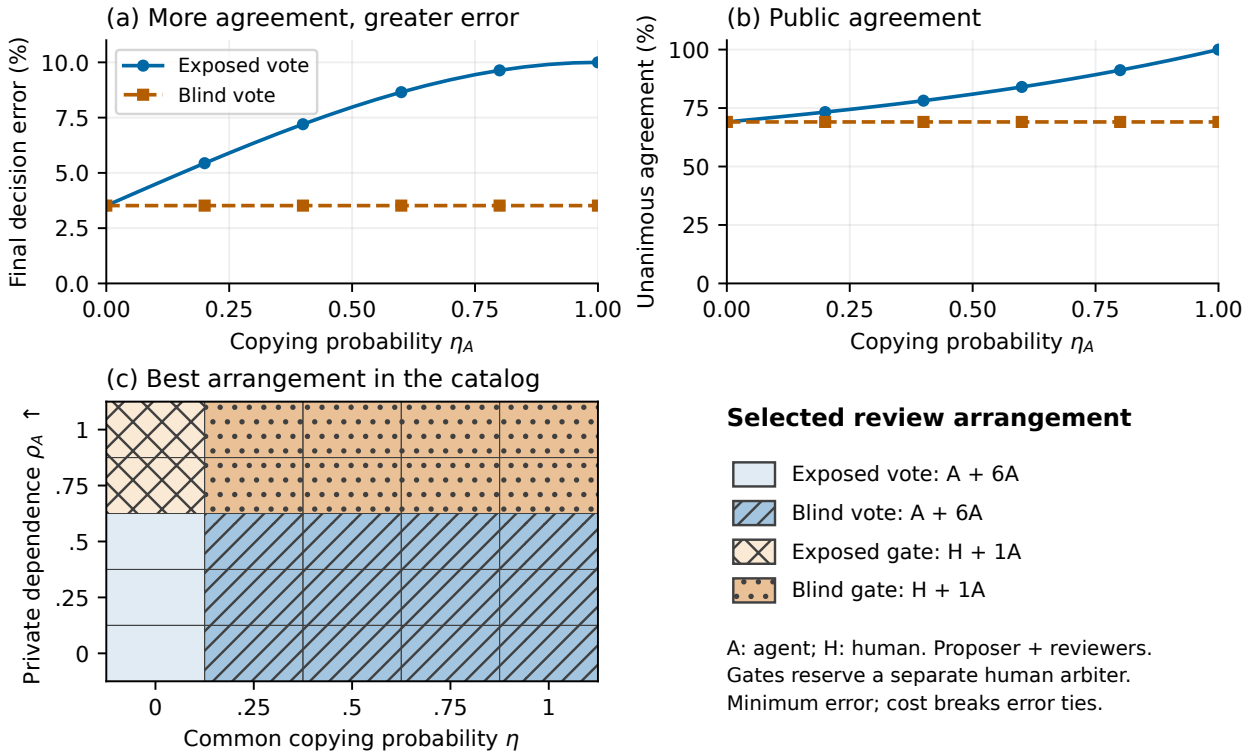


Figure 4: Study 2: private judgment, upstream exposure, and review organization. (a–b) Final majority-decision error and unanimous agreement for an agent proposer and three agent reviewers, with $q_A = .1$ and $\rho_A = .1$. Exposed reviewers copy with probability η_A; blind protocols retain private votes. The curves evaluate 21 equally spaced copying probabilities. (c) The minimum-error arrangement, with cost breaking error ties, at each dependence and copying setting, for $q_H = .2$, $q_A = .1$, and a hard task budget of three human-call units. Human and agent calls cost 1 and .25; blind commitment adds .05 per reviewer. Gates reserve a separate human arbiter. Dependence increases upward. Colors and hatching identify the selected arrangements; all values are exact expectations.

In High echo with cumulative memory, paired observation achieves mean decision error 1.77% and net value .861, compared with 9.82% and .623 for the Omit echo ablation using the same acquired records. Randomized exposure reaches 1.78% error and .861 net value. The fixed blind agent-proposer/three-reviewer baseline has higher error, 3.52%, but higher net value, .871, because it purchases no continuing learning evidence. Appendix E reports all six environments under cumulative memory; Supplementary Table S5 reports every policy–memory condition and its costs.

Across all six environments, cumulative paired observation has lower error than the fixed blind baseline, but its net value is lower in five. It spends 234 acquisition units per round, or .0570 per production task after amortization over 4096 tasks. In those five environments this expense exceeds the gain in deployment value, including production call cost; the switch charge is included separately. In Stronger humans, paired observation's net value is .773, compared with .607 for the fixed blind baseline.

In No echo, fixed exposed and fixed blind review have the same error rate, but net values of .874 and .871: blind review pays commitment overhead and one initial switch. Under cumulative memory, randomized exposure has a 56.7% optimal-choice hit rate yet the same mean error as paired observation (1.77%); its higher production call cost makes some equally accurate choices suboptimal under deployment loss. Error, regret, and net value distinguish these equally accurate but differently priced choices.

Memory effects depend on the observation policy. In Echo shift, paired observation's full-horizon net values are .858 with reset, .863 with cumulative memory, and .863 with Window 8; cumulative and Window 8 both reach final-16-round error 1.77%. For randomized exposure, the corresponding net values are .851, .852, and .857, and final-16-round errors are 2.32%, 1.84%, and 1.77%. Finite retention improves adaptation for randomized exposure, while paired observation changes little between cumulative and Window 8 memory. The value of forgetting depends on the observation procedure.

6 Implications for human–agent organizations

6.1 Comparing capability and organizational change

The specification makes an otherwise ambiguous comparison operational. A capability-only intervention changes actor profiles while keeping assignment, visibility, authority, and improvement procedures fixed. A workflow intervention changes those arrangements under a fixed procedure. A procedure intervention changes how candidate arrangements are evaluated or retained. A factorial study can compare these interventions and their interactions under the same task population and outcome criterion. The bottleneck example in section 2 supplies one elementary prediction; Study 1 shows how evidence acquisition and retention alter the value of organizational search; Study 2 shows how review arrangements generate the judgments that can become evidence.

The distinction matters when AI changes both production volume and the work of supervision. Automation can redistribute monitoring demands [6]; human and machine learning can become organizationally interdependent [42]. In Study 1, the same amount of acquired evidence has different value depending on whether the organization reuses it, discards it, or retains it after the relevant task relationship changes. Evidence retention should therefore be evaluated together with the workflow that generates it.

The supplementary audit model in Appendix D examines the earlier decision to acquire information at all. Its finite-horizon value-of-information controller can acquire little evidence under a strong prior, while Study 1 asks what happens to evidence after it has been bought. Together the models distinguish acquisition, retention, and use. Neither an audit count nor a revision count alone measures an organization's capacity to learn.

6.2 Independence and interaction order

Distinct agents can share models, training data, tools, and explanations, producing correlated errors [13]. Correlated-vote extensions of Condorcet's jury theorem analyze dependent judgments [27]; information-cascade theory and social-influence experiments examine how exposure changes the use and diversity of private information [9, 31]. Study 2 separates shared private errors from copying and shows that a two-actor public record can conceal this distinction. Recording initial judgments and reveal events makes the mechanisms distinguishable.

Human-first and agent-first interaction differ through the information available when a person commits an assessment. Private commitment can preserve an additional signal, while early advice can supply knowledge the person lacks. The outcome depends on capability, covariance, reliance, and the aggregation rule. The elementary calculation in Appendix B shows that apparent gains from exposure can equal a reweighting of private evidence. Human–AI studies of appropriate reliance and cognitive forcing motivate behavioral measurements of responses to advice [7, 12, 28]; their task and participant measures can be used to estimate how exposure changes reliance.

6.3 Preserving disagreement and separating observation from authority

Study 2 identifies information exposure and decision rights as distinct design choices. When reviewers can copy an upstream decision, public agreement combines agreement already present in private judgments with exposure-induced conformity. An organization that escalates only visible disagreements can therefore suppress its own opportunity to correct mistakes. Retaining initial assessments, their sources, and reveal order makes these mechanisms distinguishable and supports comparisons of alternative review arrangements.

A gold-labeled audit measures the performance of a review arrangement. A human override changes its final decisions. Their costs and benefits belong to different parts of the comparison: the audit informs future arrangement choices, while the override affects the current task. This distinction helps a team decide whether it needs better measurement, a different escalation rule, or a different final decision maker.

6.4 How the two studies connect

Study 1's workflow reversal and Study 2's copying shift both change the relevance of earlier observations. Their different adaptation patterns show that memory must be assessed together with the process generating those observations. The acquisition-matched control isolates evaluator replacement from a changed information path; the Omit echo ablation checks how a deliberately misspecified model interprets the same acquired record. Study 2 holds observation policies fixed, extending the operational comparison to review arrangements whose evidence is generated internally.

6.5 Validity and next empirical tests

The reference checker and public-data mapping demonstrate how the specification links decisions to evidence, permissions, versions, and costs. A comparative evaluation of the specification could measure model-building time,

agreement about missing records, and prediction accuracy when users model the same organizations with alternative notations.

Study 1 uses a finite program catalog, known stratum weights, and retained scores from trials conducted at different knowledge states. A wider catalog and adaptive weighting of historical scores would extend the acquisition-matched comparison. Concept-drift, switching-bandit, and change-point methods provide alternatives to fixed-window retention [1, 17, 18]; acquisition costs can be included in those comparisons [43].

Study 2's next tests concern when to stop paying for observation, how deployed private judgments can supply feedback, and how production volume changes the value of that feedback. Field measurements would also need to estimate human copying of agent advice and whether recording a private commitment changes subsequent behavior. These measurements would connect the error and copying assumptions to a particular review task.

For Study 1, an organizational test would randomize review-routing changes and evaluation coverage across comparable task streams, retain failed proposals, and measure escaped errors and human attention over a declared horizon. A second stage could randomize whether teams may replace the evaluation program. Such a design would separate the value of broader evidence from the value of the procedure used to discover and retain that evidence policy.

For Study 2, an organizational test would randomize upstream visibility before private commitment across comparable review tasks, while recording actor identity, initial and final judgments, independent outcome labels, and attention costs. Separately varying escalation and override rights would distinguish the value of additional evidence from the effects of final decision authority. A later stage could randomize whether teams may replace the observation protocol, thereby testing procedure revision beyond the operational arrangement choices studied here.

7 Conclusion

Recursive organization improvement asks how human–agent teams change their work and learn whether those changes helped. The proposed specification links organizational arrangements to visible evidence, decision rights, retained versions, and outcome comparisons. In Study 1, evidence reuse removes much of repeated assessment's apparent disadvantage, while obsolete evidence delays adaptation. Acquisition-matched controls leave small, uncertain differences between fixed and replaceable evaluation programs across three reversal timings. Study 2 shows how review arrangements determine which judgments become observable and how shared errors and copying change the placement of human judgment under a budget. Lower decision error can still come with a net loss from continued observation. Organizational improvement should therefore be assessed through the evidence acquired and retained; acquisition-matched comparisons can isolate the contribution of revising an evaluator.

Code and data availability

Code, protocols, trajectory results, validation records, and the public-record snapshot are available at:

`https://github.com/wizardlancet/recursive-organization-improvement`

Protocols were archived at commits `75d312e` and `fdf5ec1`; neither study was externally preregistered. Supplementary Tables S1–S5 follow the appendices. S2 summarizes final program-selection proportions; the repository contains the full 189-group conditional results and maps each table to its source. Code is licensed under MIT; the manuscript, figures, and synthetic data under CC BY 4.0. Third-party materials retain their original terms.

Use of AI tools

OpenAI Codex assisted with manuscript drafting and revision, simulation implementation, data analysis, and figure preparation.

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A Instantiating the specification: a three-PR feasibility mapping

An instance begins by declaring its population, criterion, horizon, and change boundary. It then assigns actors to roles, defines dependencies and visibility, and supplies the initial arrangement and improvement procedure. Evidence references identify versioned artifacts rather than undifferentiated transcript text. Each patch records the state it expects and the evidence its proposer could inspect. A compact contract can use the following interchange form; the referenced artifacts contain the population, protocol, scores, and retention details.

```
{
  "id": "coverage-change-1",
  "target": "evaluation.coverage_program",
  "expected_version": 0,
  "actor": "review_board",
  "transformation": {"from": "Biased", "to": "Balanced"},
  "evidence": ["program-trials-1.v1"],
  "comparison": "population-and-net-value-criterion.v1",
  "cost_ledger": "labels-outputs-transition-1.v1",
  "horizon_rounds": 8,
  "retention": "next-review-rule.v1"
}
```

The example links a proposed change to the records needed to evaluate it. The reference checker implements target, version, actor, evidence, reversibility, and admission-cost fields; the experiment protocol defines the remaining fields. A diagnostic coverage trace records sample counts, estimates, decisions, costs, and authorized program changes. Outcome equations reconstruct net value from the selected workflow and ledger.

The public mapping uses a convenience sample from `pallets/flask`, frozen on September 28, 2026. From the 100 closed PRs returned in descending creation order at retrieval, it takes the first three already merged PRs. The archive retains the candidate list, URLs, timestamps, and response hashes. Message bodies and email addresses are excluded. Actor identifiers are replaced by stable pseudonyms; the public PR links permit re-identification.

For each PR the adapter retrieves commits, reviews, and checks at its recorded head SHA, checking pagination for the subordinate lists. Table 5 summarizes the mapped records. Commits are artifact versions; opening, review, check, and merge records are events. Endpoint definitions support these mappings [19, 20]. Earlier heads, test-merge commits, off-platform activity, and historical branch protection require additional sources. The adapter records historical permission

Table 5: Public-record snapshot. Counts summarize the records available for mapping.

PR	Commit artifacts	Reviews	Head checks	Events
#6133	1	0	14	16
#6096	2	5	14	21
#6095	1	3	14	19

and actual evidence exposure as unknown. The mapping identifies the links that additional instrumentation would need to recover before a workflow-effect comparison.

B Analytical background for interaction and evaluation

B.1 Coverage and indistinguishability

Suppose two environments induce identical distributions over every history accessible to a procedure, but a specified workflow change improves the population in one and harms it in the other. Any binary sign decision based on those histories has the same output distribution in both environments. If its probability of reporting improvement is q, its two error probabilities are $1-q$ and q. Their sum is one, so at least one is at least $1/2$. This is the standard observation-equivalence argument.

A construction uses equally frequent flagged and unflagged cases. A change lowers flagged error from .20 to .10 in both environments; unflagged error falls to .10 in one and rises to .50 in the other. If only flagged labels are observed and no downstream signal arrives, evidence is identical although population risk changes from .20 to .10 or .30. With known positive inclusion probabilities q_i and accurate labels E_i, the Horvitz–Thompson estimator $N^{-1}\sum_i Z_i E_i/q_i$ is design-unbiased for finite-population risk [21]. Small inclusion probabilities can still produce high variance. Study 1 gives every cell positive coverage and examines the resulting finite-sample selection problem.

B.2 Dependent evidence and interaction order

For equal-variance errors with equicorrelation ρ, the variance of the mean is $\sigma^2[\rho+(1-\rho)/n]$, with $-1/(n-1) \le \rho \le 1$. Common bias contributes an additional squared-bias term to mean squared error. Information dependence therefore limits the gain from adding actors [13]; more actors alone need not produce proportionally more information.

Let independent initial human and agent estimation errors have variances v_H and v_A. Suppose the human sees the agent answer and replaces the private judgment by $(1-\alpha)H + \alpha A$. Equal averaging of that judgment with A has risk

$$R(\alpha) = \{(1-\alpha)^2 v_H + (1+\alpha)^2 v_A\}/4. \tag{6}$$

For $v_A < v_H$ and $0 \le \alpha \le 1$, the optimum is $\alpha^* = (v_H - v_A)/(v_H + v_A)$, giving $v_H v_A/(v_H + v_A)$. This is also the risk of optimal inverse-variance weighting of the original private judgments. Thus reweighting the private judgments can reproduce the gain over equal private averaging. With initial covariance σ_{HA}, risk gains the term $(1-\alpha^2)\sigma_{HA}/2$.

C Study 1: complete specification

C.1 Core protocol and exploratory controls

The core design specifies 48 rounds, a 2736-label ceiling for each eight-round block, three catalog programs, six arms, three memory conditions, three environments, and the allocation grid. The trial-matched control holds trial acquisition and label reuse fixed while varying whether the retained evaluation program can change; the timing checks move reversal to rounds 21 and 29. Code and data availability identifies the archived protocols.

The core has $3 \times 3 \times 6 \times 128 = 6912$ trajectories. The allocation grid uses 96 replicates for each share/trial-count combination and each discovery arm under stationary cumulative memory and reversal with Window 8. Four fixed references and two memory ablations in each environment bring this analysis to 4608 trajectories. The exploratory controls cross four environments (stationary and reversals before rounds 21, 25, 29), three memory conditions, four arms, and 128 replicates: 6144 trajectories. The three independent keyed root seeds are 920000, 930000, and 940000. Each random stream is indexed by root seed, round, acquisition stage, program, trial, and replicate. Conditions share underlying potential-label streams within each root; no policy receives unqueried labels.

C.2 Evidence update and program trials

For a template–stratum cell, cumulative memory adds each actually queried label once. Window 8 uses observations at historical rounds $t-8$ through $t-1$, plus current observations. Reset retains no earlier observations. All program trials at a review receive the same pre-review memory snapshot. Trial outputs and fresh validation labels are pooled only after program selection, then made available to the current operational screen. Validation observes only the selected template, while screening observes all templates except for the second phase of the rejection comparator. A label from a rejected program remains usable if its source and population are appropriate. The pooling ablation withholds all program-trial labels from workflow memory while retaining program scores.

Each trial score is $1 - 3\widehat{r}^{\,\mathrm{val}} - .22N^{\mathrm{screen}}/4096$, where validation uses equal counts across the two strata. A program's score is the mean of its trial means retained under the memory rule. Each review has the same trial count within a configuration. Ties favor Biased, then Balanced, then Neyman. Program-score memory can be disabled independently of workflow memory. Retained scores average trials performed under different historical knowledge states.

For block budget $B = 2736$, program share f, three programs, and q trials per program, a trial receives $b = \lfloor \lfloor fB \rfloor / (3q) \rfloor$ label units. Validation obtains $v = \max\{1, \lfloor \lfloor .2b \rfloor / 2 \rfloor\}$ labels per stratum; screening receives $b - 2v$ total units. The base $(f, q) = (.2, 1)$ therefore allows 182 units per trial, with 18 validation labels per stratum and 146 units for screening. Actual counts can be smaller after integer allocation. Let M_i be the actual number of trial labels for replicate i at a block's start. Its subsequent per-round operational allowance is $\lfloor (B - M_i)/8 \rfloor$. Discover once has $M_i = 0$ after block 1. This rule keeps the budget fixed when trial count or program share changes.

C.3 Within-template and between-template allocation

For the catalog programs, let u be one template's share of the operational or trial allowance. Balanced obtains $\lfloor u/2 \rfloor$ labels in each stratum. Biased obtains $\max\{1, \lfloor .95u \rfloor\}$ and $\max\{1, \lfloor .05u \rfloor\}$. All tested allocations fit the allowance. Neyman first obtains $m = \min\{2, \max\{1, \lfloor u/4 \rfloor\}\}$ labels per cell. Using retained and pilot data, it estimates p_s, sets $v_s = p_s(1-p_s)$, and allocates

$$n_s = m + \left\lfloor \frac{(u-2m)\sqrt{v_s}}{\sum_\ell \sqrt{v_\ell}} \right\rfloor. \tag{7}$$

Pilot labels enter the posterior once. Equal label costs make this the equal-cost stratified allocation. Catalog selection breaks posterior ties in template order standard, specialized, broad.

The successive-rejection comparator treats one pair of stratum labels as a bounded loss with mean equal to population risk. With n affordable pairs and $K = 3$, it uses the schedule of Audibert et al. [5]:

$$\overline{\log}K = \frac{1}{2} + \sum_{j=2}^{K} \frac{1}{j}, \qquad n_k = \left\lceil \frac{n - K}{(\overline{\log}K)(K + 1 - k)} \right\rceil, \quad k = 1, 2. \tag{8}$$

All three templates receive n_1 pairs; the worst posterior-mean template is removed, and the remaining two reach n_2 pairs each. Elimination ties remove the lowest indexed template; final ties use the lowest surviving index. Historical evidence enters the posterior under the memory conditions. Thus the implementation uses the published rejection schedule with declared posterior ranking and deterministic ties. At the fixed allowance of 342 labels it uses 336 labels. SR is a fixed between-template comparator, separate from the three-program discovery catalog.

C.4 Outcome accounting, checks, and uncertainty

For deployed template k_t with true population risk $r_{k_t,t}$, total queried labels Q_t, and program-change indicator S_t, the external score is

$$V_t = 1 - 3r_{k_t,t} - .02 - \{.22Q_t + .50S_t\}/4096. \tag{9}$$

Each queried label corresponds to a generated evaluation output, so .22 includes .20 labeling and .02 generation. The separate .02 term prices each production output. Potential-label arrays define the common random streams; actors observe queried entries. The ledger includes rejected-program trials.

The simulator checks every block ceiling and the identity $V_t = 1 - 3r_t^* - .02 - \text{regret}_t - \text{expense}_t$, where r_t^* is the best available risk. An independent scalar reconstruction checks 36 fixed-Balanced trajectories across the three environments and memory conditions to absolute tolerance 10^{-12}. Separate fixtures check the lookback boundary, rejection allocation, and selection of a known best template. The patch checker enforces target authority and state version for program changes.

Reported half-widths are $1.96s/\sqrt{N}$ across independent replicate trajectories. Paired intervals use within-replicate differences under the common streams. The uniform fixed benchmark averages the conditional mean values of Biased, Balanced, and Neyman under an initial uniform program prior; its paired contrast integrates over this discrete prior analytically. Intervals are unadjusted for multiple comparisons.

Table 6 reports all core net-value means. Supplementary Table S1 reports their Monte Carlo intervals, costs, harmful adoption, selection regret, and late outcomes; replicate-level data accompany the reproducibility materials.

Table 6: Complete core net-value means. B: Biased; Bal: Balanced; N: Neyman; Mix: uniform fixed-program mixture; SR: successive rejection; O: discover once; R: repeated discovery. Each cell uses 128 trajectories. Monte Carlo intervals and additional outcomes are reported in Supplementary Table S1.

Environment	Memory	B	Bal	N	SR	O	R	Mix
Stationary harm	Reset	0.3967	0.4522	0.4539	0.4540	0.4366	0.4352	0.4343
Stationary harm	Cumulative	0.4692	0.4801	0.4805	0.4806	0.4793	0.4800	0.4766
Stationary harm	Window 8	0.4594	0.4792	0.4796	0.4800	0.4754	0.4779	0.4727
Workflow reversal	Reset	0.5260	0.5828	0.5830	0.5844	0.5644	0.5628	0.5639
Workflow reversal	Cumulative	0.4721	0.4803	0.4805	0.4829	0.4796	0.4835	0.4776
Workflow reversal	Window 8	0.5609	0.5758	0.5760	0.5786	0.5723	0.5821	0.5709
Uniform gain	Reset	0.6617	0.7141	0.7132	0.7162	0.6962	0.6941	0.6963
Uniform gain	Cumulative	0.7184	0.7215	0.7216	0.7218	0.7217	0.7219	0.7205
Uniform gain	Window 8	0.7166	0.7215	0.7216	0.7218	0.7210	0.7219	0.7199

Table 7 reports the allocation grid. Supplementary Table S2 summarizes the retained-program proportions at each rule's final scheduled assessment; the repository contains the conditional outcomes for every assessment.

Table 7: Repeated discovery minus discover-once at fixed total budget. All program shares and trial counts are shown. Entries are paired differences ± 95% Monte Carlo half-width; 96 fresh replicates per condition.

Program share	Trials/program	Stationary, cumulative	Reversal, Window 8
0.2	1	+0.0012 ± 0.0012	+0.0071 ± 0.0024
0.2	3	+0.0019 ± 0.0020	+0.0053 ± 0.0023
0.2	6	+0.0006 ± 0.0008	+0.0062 ± 0.0020
0.4	1	+0.0002 ± 0.0004	+0.0105 ± 0.0025
0.4	3	+0.0002 ± 0.0001	+0.0098 ± 0.0026
0.4	6	+0.0002 ± 0.0001	+0.0080 ± 0.0019
0.6	1	+0.0007 ± 0.0011	+0.0175 ± 0.0021
0.6	3	+0.0001 ± 0.0001	+0.0158 ± 0.0022
0.6	6	+0.0002 ± 0.0002	+0.0134 ± 0.0015

Table 8: Repeated discovery: retained-program percentages and subsequent eight-round outcomes conditional on retaining Balanced. Stationary and uniform gain use cumulative memory; reversal uses Window 8. B/Bal/N are Biased/Balanced/Neyman; n_{Bal} is the number retaining Balanced among 128 replicates. Harm is the percentage of rounds deploying a template worse than standard.

Environment	Review	B (%)	Bal (%)	N (%)	n_{Bal}	Net given Bal	Harm (%)
Stationary	1	27.3	37.5	35.2	48	0.4747	0.00
Stationary	9	26.6	35.9	37.5	46	0.4819	0.00
Stationary	17	33.6	34.4	32.0	44	0.4819	0.00
Stationary	25	33.6	35.9	30.5	46	0.4819	0.00
Stationary	33	35.9	32.0	32.0	41	0.4819	0.00
Stationary	41	38.3	31.2	30.5	40	0.4819	0.00
Reversal	1	27.3	37.5	35.2	48	0.4747	0.00
Reversal	9	26.6	35.9	37.5	46	0.4783	0.00
Reversal	17	43.8	27.3	28.9	35	0.4819	0.00
Reversal	25	39.8	30.5	29.7	39	0.6188	0.00
Reversal	33	43.8	28.1	28.1	36	0.7219	0.00
Reversal	41	47.7	20.3	32.0	26	0.7219	0.00
Uniform gain	1	35.9	27.3	36.7	35	0.7219	0.00
Uniform gain	9	26.6	30.5	43.0	39	0.7219	0.00
Uniform gain	17	25.8	29.7	44.5	38	0.7219	0.00
Uniform gain	25	28.1	30.5	41.4	39	0.7219	0.00
Uniform gain	33	28.1	28.1	43.8	36	0.7219	0.00
Uniform gain	41	30.5	25.8	43.8	33	0.7219	0.00

D Supplementary audit-control model

The supplementary audit model isolates longitudinal evidence acquisition under a fixed controller. It contains 160 periods of 32 tasks. Route A has base value 1 and error probability .04 (stationary) or alternating $.04/.25$ blocks of 40 periods; route B has base value .82 and error probability .08. An error costs 3, an audit costs .20, and an audit-rate change costs .50 per batch. Only audits reveal A's errors, available next period; executing B can still be paired with acquiring an A audit. Observations update a sliding-window Beta estimate. The base prior is $\mathrm{Beta}(1, 9)$ and window 12.

The payoff-optimal routing threshold is .14, since $1 - 3p = .82 - 3(.08) = .58$. The heuristic adaptive controller audits at rate .40 when posterior $P(p > .14)$ lies between .10 and .90, and at .025 otherwise. Periodic dense auditing uses .40 for the first three periods of each 12-period cycle and .025 otherwise. The base audit comparison uses 400 seeds from 20260928, common across policies. Stationary net values are .86089 for fixed .025, .86395 for fixed .05, .85841 for fixed .10, and .87976 for no auditing. No auditing retains A under the base prior, the optimal route in that stationary case; audits cannot repair current outputs.

With slow shifts, adaptive auditing exceeds fixed .10 by .00857 per task (paired interval .00676–.01038). A separate 200-seed grid, beginning at 20360928, crosses windows 6, 12, 24 with stationary, 40-period, and 10-period regimes. It includes no audit, fixed $.05/.10/.40$, periodic dense, and adaptive policies. Supplementary Table S3 reports all 54 combinations of environment, memory window, and audit policy.

D.1 Finite-horizon EVSI controller

The finite-horizon expected-value-of-sample-information (EVSI) controller evaluates acquiring $m \in \{0, 1, 4, 16\}$ labels. Given current Beta parameters (a, b), a beta-binomial predictive distribution over k observed errors gives posterior mean $(a + k)/(a + b + m)$. Let $L_\theta(p)$ be $1 - 3p$ if $p \le \theta$ and .58 otherwise. The controller maximizes

$$h\{\mathbb{E}_k[L_\theta((a + k)/(a + b + m))] - L_\theta(a/(a + b))\} - \frac{\left[(.20 + g_c I_B)m + .50 I_{\text{change}}\right]}{32}. \tag{10}$$

I_{change} equals 1 when the audit rate changes and 0 otherwise. Both costs in the bracket are subtracted. I_B indicates executing B; g_c is the cost of generating a counterfactual A output to audit. The predictive sum is exact under the current Beta model. The controller approximates future value with horizon h and a sliding window, so it is a finite-horizon EVSI heuristic under drift.

Development fixes the prior at $\text{Beta}(1, 9)$, the threshold at .14, and $g_c = 0$. Seeds 710000–710063 choose h from $1, 4, 8, 16$ using stationary and slow-shift mean value; $h = 16$ is selected. Testing uses 96 independent seeds 810000–810095, three priors, thresholds .14/.16, generation costs $g_c = 0/.20$, two environments, and three policies: 6912 trajectories. Threshold .16 is a misspecified routing-rule sensitivity under unchanged payoffs. Nonzero g_c charges selectively generated A outputs while B is executed; it is distinct from a common cost charged for producing every candidate under every policy.

Table 9 reports all prior/threshold means at $g_c = 0$. Supplementary Table S4 reports both candidate-generation cost conditions, marginal Monte Carlo intervals, and paired policy contrasts. EVSI's performance depends strongly on the prior: with no initial feedback, information acquisition itself can stall. Four regression cases match a reference scalar implementation to 10^{-12}.

Table 9: Supplementary audit sensitivity at zero additional candidate-generation cost. Columns give net-value means over 96 trajectories; F: fixed .10, A: adaptive heuristic, E: EVSI. Uncertainty estimates and results at $g_c = .20$ are reported in Supplementary Table S4.

		Stationary			Slow shifts		
Prior $(a : b)$	Threshold	F	A	E	F	A	E
1:1	0.14	0.8479	0.8530	0.8338	0.6616	0.6753	0.6758
1:1	0.16	0.8523	0.8571	0.8342	0.6597	0.6729	0.6757
1:9	0.14	0.8581	0.8649	0.8656	0.6620	0.6706	0.6599
1:9	0.16	0.8592	0.8686	0.8660	0.6515	0.6642	0.6526
1:19	0.14	0.8595	0.8720	0.8795	0.6468	0.6571	0.5673
1:19	0.16	0.8595	0.8735	0.8795	0.6263	0.6396	0.5673

E Study 2: complete specification

E.1 Private judgments, exposure, and decision rights

The review and learning protocols below define the actor error law, observation access, candidate arrangements, costs, and retention rules used in Study 2.

A task has balanced binary truth $y \in \{0, 1\}$. A private judgment equals y unless its actor makes an error. Human errors are independent Bernoulli draws with probability q_H, conditional on y, and independent of the agent group. For each task, draw a shared-mode indicator with probability ρ_A. In shared mode, all agent errors equal one Bernoulli draw with probability q_A; otherwise each agent draws an independent Bernoulli error with that probability. Tasks are independent. An agent proposer belongs to the same group as its reviewers. This mixture preserves marginal agent error q_A and gives pairwise error correlation ρ_A; the full mixture law determines collective error.

Let d_0 be the displayed upstream judgment and d_i^{priv} reviewer i's private judgment. An exposed reviewer reports

$$d_i^{\text{pub}} = (1 - C_i)d_i^{\text{priv}} + C_i d_0, \qquad C_i \sim \text{Bernoulli}(\eta_s), \quad s \in \{H, A\}. \tag{11}$$

Here η_s is the probability that an agent reviewer copies an upstream decision from source s; s identifies a human or agent upstream actor. Copy indicators are independent of private errors and of one another. The copying rule selects either the private judgment or the displayed upstream judgment. Blind protocols use private commitments for aggregation or escalation, so changing η_s does not change their decisions. In serial review, the immediately preceding review replaces d_0 at each step, and all sources after the first step are agents.

Table 10 defines the six protocol families and two single-actor baselines. A gate retains the proposal only if all relevant reviews agree with it; otherwise a separate human makes the final decision. The human arbiter has error probability q_H and is independent of the proposer and reviewers. A human proposer and a human arbiter therefore require two distinct humans. A production audit grants the arbiter override authority, whereas a gold label supplies correct measurement for learning. These rules separate private information, public disagreement, and final authority. For example, the

Table 10: Review protocols and final decision rights. Voting includes the proposer and all reviewers; a tied vote uses a fair random choice. All reviewers are agents. Gates call an additional human only on the indicated cases.

Protocol	Evidence used	Final decision
Exposed vote	Reviews after seeing the proposal	Majority vote.
Blind vote	Private commitments	Majority vote.
Exposed disagreement gate	Public reviews versus proposal	Human arbitration on any disagreement.
Blind disagreement gate	Private reviews versus proposal	Human arbitration on any disagreement.
Gate with random arbitration	Public reviews versus proposal	Arbitrate all disagreements and a random 25% of unanimous cases.
Serial last-reviewer decision	Each reviewer sees only its immediate predecessor	Last review; no collective vote.
Single actor	One human or agent private judgment	That actor's judgment.

change in error from randomly overriding a unanimous case is $q_H - P(d_0 \neq y \mid \text{unanimous})$. An independent human override helps only when this difference is negative.

The exact search crosses $q_H \in \{.1, .2, .3\}$, $q_A \in \{.05, .1, .2, .3\}$, $\rho_A \in \{0, .25, .5, .75, 1\}$ and $\eta \in \{0, .25, .5, .75, 1\}$ with $\eta_H = \eta_A = \eta$, human-to-agent call-cost ratios $2, 4, 8$, blind overheads $0, .05, .2$ per reviewer, task budgets $1, 2, 3$, and hard versus expected budget constraints. Human calls define one cost unit. A hard budget reserves every potentially called actor; an expected budget constrains the mean cost including escalation probabilities. The catalog has $2 \times 7 \times 6 + 2 = 86$ arrangements: two proposer types, one to seven reviewers, six protocols, and two single-actor baselines. Exact finite-state probabilities identify the minimum-error feasible arrangement in each of 16,200 cells. Errors within 10^{-12} count as ties; lower expected cost breaks error ties. Remaining ties use catalog order: human then agent baseline, then human/agent proposer, increasing reviewer count, and the six protocols in table order. The deployment-loss optimum is recorded separately.

The fixed-roster comparison retains all 25,200 combinations of the accuracy, dependence and exposure grid, two proposer types, one to seven reviewers, and six protocols. A separate source-sensitivity grid uses $\eta_H, \eta_A \in \{0, .2, .5, .8, 1\}$, $\rho_A \in \{0, .5, .9\}$, $q_H = .2$, and $q_A = .1$, giving 6,450 arrangement evaluations. The curves in figure 4a–b evaluate 21 equally spaced exposure values.

E.2 Paid observations and retained evidence

An observer can learn individual ability and copying only from the information it buys. For an agent proposer and one equally capable agent reviewer,

$$P(d_1^{\text{pub}} \neq d_0) = 2q_A(1 - q_A)(1 - \rho_A)(1 - \eta_A).$$

Consequently, at the same q_A, $(\rho_A, \eta_A) = (.8, 0)$ and $(0, .8)$ give the same joint distribution of public votes and truth in this two-actor case, but different responses to blinding. Private observations or randomized visibility can distinguish the mechanisms.

Four fixed observation policies select among 58 arrangements with at most five reviewers and hard task cost at most three. Human, agent, gold-label, and per-reviewer blind-commitment costs are $1, .25, 1, .05$, respectively. Each round permits at most 240 acquisition-cost units, hence 1920 per eight-round block, with no borrowing. These cost units differ from the label counts used in Appendix C. All current observations are acquired and charged before the selected arrangement is deployed; production outcomes do not provide free learning feedback.

Paired observation records two human and six agent private judgments, ten additional exposed agent responses (five for each proposer type), a gold label, and six private-commitment overheads on each task. Its cost is $2+6(.25)+10(.25)+1+6(.05) = 7.3$, allowing 32 tasks and costing 233.6 per round. The same task supplies pre- and post-exposure judgments. Among reviews that initially disagree with a source, the fraction switching to that source estimates η_s. Recording a private commitment leaves the later copying probability unchanged in this model.

Randomized exposure acquires 50 tasks, randomized in balanced cells crossing human/agent source and blind/exposed review. Each task purchases two human responses, six agent responses, and a gold label. Blind tasks additionally pay six commitment overheads. The four cells have 12 or 13 tasks, with 25 blind tasks in total, costing $25(4.8)+25(4.5) =$

232.5 per round. Private ability and dependence use only blind tasks. For each source, the estimate of copying is one minus exposed disagreement divided by blind disagreement, clipped to $[0, 1]$. This design uses disjoint blind and exposed tasks and assumes exchangeability across their randomized assignments.

Both model-based policies estimate q_H and q_A from gold-labeled private errors with one error and one success pseudocount, then clip estimates to $[.001, .499]$. Agent dependence is the private pairwise error moment minus squared raw error frequency, divided by the raw Bernoulli variance and clipped to $[0, 1]$; zero estimated variance gives zero dependence. Paired copying estimates use half an event in each outcome as pseudocounts. Randomized disagreement rates likewise use half-event pseudocounts before taking their ratio. Each policy pools the requisite counts over its assigned memory interval and predicts every feasible arrangement's error and call cost under the known model family. The true parameters are used only by the external scorer and catalog benchmark.

Omit echo ablation purchases exactly the paired observations but fixes $\eta_H = \eta_A = 0$ when predicting arrangements. *Final-outcome sampling* instead buys an arrangement's executed final outcome and a gold label, cycling through candidates. At zero-based round r, trial j uses catalog index $(83r + j) \bmod 58$. A trial reserves its worst possible call cost plus one gold label; actual calls and that label are charged. It stops before exceeding the ceiling. For a candidate with n retained observations, e errors, total call cost c, and worst call cost $c_{\max}$, its score is $3(1+e)/(2+n) + .02(c + c_{\max})/(1+n)$. Thus unobserved candidates also have defined scores.

Reset retains current-round counts; cumulative memory retains all acquired counts; Window 8 retains the current round and eight preceding rounds. Observation policies and memory rules are assigned research treatments and remain fixed. Each selected arrangement changes operational visibility, aggregation, or authority under the assigned procedure.

E.3 Evaluation, reporting, and validation

The six learning environments in table 11 isolate copying, private dependence, source asymmetry, and relative actor ability. The copying shift occurs before acquisition in round 25. Each environment crosses four observation policies and three memories, plus two fixed baselines, with 128 replicate trajectories of 48 rounds: $6(4 \times 3 + 2)128 = 10{,}752$ trajectories. Fixed baselines use an agent proposer and three agent reviewers, either exposed or blind, and buy no learning evidence. The model-based learner selects the arrangement minimizing estimated deployment loss $3e + .02c$,

Table 11: Study 2 environments. In the environment and policy names, echo means copying. All other costs and resource ceilings are held fixed as specified in Appendix E.

Environment	q_H	q_A	ρ_A	η_H	η_A	Change
No echo	.2	.1	.1	0	0	None
High echo	.2	.1	.1	.8	.8	None
Both dependences	.2	.1	.8	.8	.8	None
Echo shift	.2	.1	.1	0	0	Both η_s become .8 at round 25
Source asymmetry	.2	.1	.2	.9	.1	None
Stronger humans	.1	.2	.2	.8	.8	None

where e is final error probability and c is expected production call cost. Ties use the filtered catalog order. Selection regret compares this loss with the feasible catalog minimum under the true environment parameters; an optimal-choice hit accepts every loss within 10^{-10} of that minimum. This objective differs from the error-first grid search and excludes acquisition and switching costs. Net value additionally charges those costs:

$$V_t^{\text{actor}} = 1 - 3e_t - .02c_t - \frac{A_t + .5S_t}{4096}, \tag{12}$$

where A_t is purchased-observation cost and S_t indicates a change from the previous arrangement. All conditions start from the exposed agent-proposer/three-reviewer arrangement; the fixed blind baseline therefore pays one initial switch. The denominator 4096 is the production-volume scale over which observation and switching costs are amortized. The scorer uses expected production outcomes.

Learning uses root seed 2026100204, keyed by environment and round, with independent replicate rows and common potential-task panels across policies. Only purchased entries are revealed. Full-horizon and final-16-round summaries average within a trajectory before computing means and $1.96s/\sqrt{128}$ Monte Carlo half-widths across trajectories. Paired comparisons use within-replicate differences. The fixed baselines have deterministic expected-outcome scores.

Table 12 reports all six environments under cumulative memory; Supplementary Table S5 gives all 84 policy–memory–environment conditions, late outcomes, costs, regret, and paired comparisons. Error and net value are reported together because better selection need not repay continuing acquisition costs.

Table 12: Study 2: selection error and net value under paid observation. All six environments are shown; learning policies use cumulative memory. Entries are means ± 95% Monte Carlo half-widths over 128 trajectories of 48 rounds. Fixed baselines have exact expected-outcome scores under the model. Error is a percentage; hit is the percentage of rounds attaining a deployment-loss minimum within 10^{-10}.

Environment	Observation / fixed rule	Error (%)	Net value	Hit (%)
No echo	Paired	1.772 ± 0.002	0.86475 ± 0.00008	98.70
	Randomized	1.772 ± 0.003	0.86333 ± 0.00026	56.67
	Omit echo	1.772 ± 0.002	0.86480 ± 0.00008	99.97
	Final only	4.919 ± 0.197	0.78045 ± 0.00563	14.47
	Fixed exposed	3.520	0.87440	0.00
	Fixed blind	3.520	0.87140	0.00
High echo	Paired	1.773 ± 0.003	0.86077 ± 0.00009	99.95
	Randomized	1.782 ± 0.008	0.86078 ± 0.00024	99.82
	Omit echo	9.825 ± 0.004	0.62323 ± 0.00013	0.00
	Final only	5.983 ± 0.217	0.74760 ± 0.00640	10.89
	Fixed exposed	9.637	0.69089	0.00
	Fixed blind	3.520	0.87140	0.00
Both dependences	Paired	7.423 ± 0.049	0.69099 ± 0.00105	82.29
	Randomized	7.498 ± 0.063	0.68945 ± 0.00135	77.10
	Omit echo	15.939 ± 0.317	0.43965 ± 0.00969	0.00
	Final only	9.728 ± 0.136	0.63630 ± 0.00404	5.49
	Fixed exposed	9.919	0.68242	0.00
	Fixed blind	8.560	0.72020	0.00
Echo shift	Paired	1.779 ± 0.006	0.86255 ± 0.00018	98.54
	Randomized	2.113 ± 0.083	0.85206 ± 0.00238	72.40
	Omit echo	5.797 ± 0.002	0.74404 ± 0.00008	49.97
	Final only	5.648 ± 0.217	0.75770 ± 0.00621	11.20
	Fixed exposed	6.579	0.78264	0.00
	Fixed blind	3.520	0.87140	0.00
Source asymmetry	Paired	2.693 ± 0.004	0.83318 ± 0.00013	99.76
	Randomized	2.936 ± 0.046	0.82728 ± 0.00116	70.91
	Omit echo	3.546 ± 0.015	0.81160 ± 0.00044	0.00
	Final only	6.352 ± 0.191	0.73785 ± 0.00535	5.34
	Fixed exposed	5.101	0.82697	0.00
	Fixed blind	4.240	0.84980	0.00
Stronger humans	Paired	4.623 ± 0.009	0.77306 ± 0.00028	99.37
	Randomized	4.639 ± 0.014	0.77286 ± 0.00041	98.96
	Omit echo	8.969 ± 0.024	0.64781 ± 0.00072	0.00
	Final only	9.522 ± 0.263	0.63271 ± 0.00748	10.04
	Fixed exposed	19.570	0.39290	0.00
	Fixed blind	12.320	0.60740	0.00

The error and cost comparisons are discussed in Section 5.4.

An independent bit-level enumeration checks 144 arrangement–environment cases to maximum absolute difference 1.11×10^{-15}. A second implementation simulates 150,000 tasks in each of four independent environments and checks 344 arrangements on the shared panels; each metric must agree within seven estimated standard errors plus 2×10^{-5}. The separate validation seed is 2026100205. Further checks cover the memory lookback boundary, copying endpoints, both resource ceilings, and the net-value identity. Exact enumeration results have no sampling interval; the Monte Carlo comparison checks the enumeration against a separately implemented sampler.

Supplementary Tables

Tables S1–S5 report the simulation results and supplementary audit controls. Uncertainty entries are 95% Monte Carlo half-widths across independent trajectories. Cost units and reporting horizons are defined separately for the two studies.

Table S1: Study 1: evidence acquisition and retention

Source: `results/learning_summary.csv`; `results/learning_contrasts.csv`.

Core design: 54 conditions, each with 128 independent replicate trajectories of 48 rounds. Window 8 includes eight previous rounds plus current observations. Net value, regret and expenses are per production task. Harm is the percentage of rounds selecting a template worse than standard. Late net value covers the final eight rounds. Panel C separates core and exploratory control contrasts; intervals use paired replicate differences.

A. Net value, harmful adoption, and selection regret

Environment	Memory	Arm (SR = Successive rejection)	Net value	Harm (%)	Regret
Stationary	Reset	Biased	0.39666 ± 0.00278	15.755 ± 0.800	0.08514 ± 0.00278
Stationary	Reset	Balanced	0.45224 ± 0.00123	0.537 ± 0.177	0.02939 ± 0.00123
Stationary	Reset	Neyman	0.45389 ± 0.00118	0.521 ± 0.175	0.02781 ± 0.00118
Stationary	Reset	SR	0.45400 ± 0.00128	0.505 ± 0.174	0.02795 ± 0.00128
Stationary	Reset	Once	0.43664 ± 0.00476	4.704 ± 1.256	0.04509 ± 0.00476
Stationary	Reset	Repeated	0.43522 ± 0.00261	4.980 ± 0.718	0.04667 ± 0.00261
Stationary	Cumulative	Biased	0.46917 ± 0.00267	1.009 ± 0.376	0.01263 ± 0.00267
Stationary	Cumulative	Balanced	0.48007 ± 0.00050	0.000 ± 0.000	0.00156 ± 0.00050
Stationary	Cumulative	Neyman	0.48046 ± 0.00048	0.000 ± 0.000	0.00133 ± 0.00048
Stationary	Cumulative	SR	0.48059 ± 0.00054	0.000 ± 0.000	0.00137 ± 0.00054
Stationary	Cumulative	Once	0.47928 ± 0.00124	0.016 ± 0.032	0.00248 ± 0.00124
Stationary	Cumulative	Repeated	0.48000 ± 0.00080	0.016 ± 0.032	0.00191 ± 0.00080
Stationary	Window 8	Biased	0.45936 ± 0.00284	1.514 ± 0.496	0.02244 ± 0.00284
Stationary	Window 8	Balanced	0.47921 ± 0.00073	0.000 ± 0.000	0.00242 ± 0.00073
Stationary	Window 8	Neyman	0.47962 ± 0.00071	0.000 ± 0.000	0.00217 ± 0.00071
Stationary	Window 8	SR	0.48002 ± 0.00069	0.000 ± 0.000	0.00193 ± 0.00069
Stationary	Window 8	Once	0.47537 ± 0.00194	0.309 ± 0.280	0.00639 ± 0.00194
Stationary	Window 8	Repeated	0.47790 ± 0.00118	0.016 ± 0.032	0.00400 ± 0.00118
Reversal	Reset	Biased	0.52595 ± 0.00282	8.203 ± 0.594	0.07584 ± 0.00282
Reversal	Reset	Balanced	0.58280 ± 0.00128	0.244 ± 0.117	0.01883 ± 0.00128
Reversal	Reset	Neyman	0.58300 ± 0.00121	0.244 ± 0.117	0.01870 ± 0.00121
Reversal	Reset	SR	0.58435 ± 0.00115	0.244 ± 0.117	0.01760 ± 0.00115
Reversal	Reset	Once	0.56442 ± 0.00519	2.702 ± 0.754	0.03731 ± 0.00520
Reversal	Reset	Repeated	0.56282 ± 0.00253	2.523 ± 0.522	0.03907 ± 0.00253
Reversal	Cumulative	Biased	0.47211 ± 0.00288	1.009 ± 0.376	0.12968 ± 0.00288
Reversal	Cumulative	Balanced	0.48026 ± 0.00060	0.000 ± 0.000	0.12137 ± 0.00060
Reversal	Cumulative	Neyman	0.48054 ± 0.00051	0.000 ± 0.000	0.12125 ± 0.00051
Reversal	Cumulative	SR	0.48285 ± 0.00131	0.000 ± 0.000	0.11910 ± 0.00131
Reversal	Cumulative	Once	0.47963 ± 0.00134	0.016 ± 0.032	0.12212 ± 0.00134
Reversal	Cumulative	Repeated	0.48347 ± 0.00235	0.016 ± 0.032	0.11843 ± 0.00235
Reversal	Window 8	Biased	0.56094 ± 0.00291	1.270 ± 0.460	0.04085 ± 0.00291
Reversal	Window 8	Balanced	0.57583 ± 0.00089	0.000 ± 0.000	0.02580 ± 0.00089
Reversal	Window 8	Neyman	0.57601 ± 0.00081	0.000 ± 0.000	0.02578 ± 0.00081
Reversal	Window 8	SR	0.57859 ± 0.00096	0.000 ± 0.000	0.02336 ± 0.00096
Reversal	Window 8	Once	0.57231 ± 0.00206	0.212 ± 0.247	0.02945 ± 0.00207
Reversal	Window 8	Repeated	0.58206 ± 0.00231	0.016 ± 0.032	0.01984 ± 0.00231
Uniform gain	Reset	Biased	0.66166 ± 0.00295	0.000 ± 0.000	0.06014 ± 0.00295
Uniform gain	Reset	Balanced	0.71411 ± 0.00111	0.000 ± 0.000	0.00752 ± 0.00111
Uniform gain	Reset	Neyman	0.71317 ± 0.00114	0.000 ± 0.000	0.00852 ± 0.00114
Uniform gain	Reset	SR	0.71619 ± 0.00097	0.000 ± 0.000	0.00576 ± 0.00097

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Environment	Memory	Arm (SR = Successive rejection)	Net value	Harm (%)	Regret
Uniform gain	Reset	Once	0.69615 ± 0.00458	0.000 ± 0.000	0.02559 ± 0.00459
Uniform gain	Reset	Repeated	0.69412 ± 0.00234	0.000 ± 0.000	0.02777 ± 0.00234
Uniform gain	Cumulative	Biased	0.71837 ± 0.00126	0.000 ± 0.000	0.00342 ± 0.00126
Uniform gain	Cumulative	Balanced	0.72147 ± 0.00019	0.000 ± 0.000	0.00016 ± 0.00019
Uniform gain	Cumulative	Neyman	0.72159 ± 0.00025	0.000 ± 0.000	0.00020 ± 0.00025
Uniform gain	Cumulative	SR	0.72184 ± 0.00017	0.000 ± 0.000	0.00012 ± 0.00017
Uniform gain	Cumulative	Once	0.72173 ± 0.00008	0.000 ± 0.000	0.00004 ± 0.00008
Uniform gain	Cumulative	Repeated	0.72187 ± 0.00008	0.000 ± 0.000	0.00004 ± 0.00008
Uniform gain	Window 8	Biased	0.71662 ± 0.00165	0.000 ± 0.000	0.00518 ± 0.00165
Uniform gain	Window 8	Balanced	0.72147 ± 0.00019	0.000 ± 0.000	0.00016 ± 0.00019
Uniform gain	Window 8	Neyman	0.72159 ± 0.00025	0.000 ± 0.000	0.00020 ± 0.00025
Uniform gain	Window 8	SR	0.72184 ± 0.00017	0.000 ± 0.000	0.00012 ± 0.00017
Uniform gain	Window 8	Once	0.72101 ± 0.00080	0.000 ± 0.000	0.00076 ± 0.00080
Uniform gain	Window 8	Repeated	0.72186 ± 0.00008	0.000 ± 0.000	0.00004 ± 0.00008

B. Evaluation expense, program-trial expense, and late outcome

Environment	Memory	Arm (SR = Successive rejection)	Expense	Trial expense	Late net
Stationary	Reset	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.39935 ± 0.00766
Stationary	Reset	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.45327 ± 0.00317
Stationary	Reset	Neyman	0.01830 ± 0.00000	0.00000 ± 0.00000	0.45577 ± 0.00305
Stationary	Reset	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.45670 ± 0.00320
Stationary	Reset	Once	0.01827 ± 0.00001	0.00060 ± 0.00000	0.44309 ± 0.00568
Stationary	Reset	Repeated	0.01811 ± 0.00000	0.00361 ± 0.00000	0.43967 ± 0.00566
Stationary	Cumulative	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.48062 ± 0.00186
Stationary	Cumulative	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.48163 ± 0.00000
Stationary	Cumulative	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.48179 ± 0.00000
Stationary	Cumulative	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.48195 ± 0.00000
Stationary	Cumulative	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.48173 ± 0.00001
Stationary	Cumulative	Repeated	0.01809 ± 0.00000	0.00359 ± 0.00000	0.48191 ± 0.00000
Stationary	Window 8	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.46571 ± 0.00562
Stationary	Window 8	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.47976 ± 0.00216
Stationary	Window 8	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.47992 ± 0.00211
Stationary	Window 8	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.48066 ± 0.00185
Stationary	Window 8	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.47719 ± 0.00265
Stationary	Window 8	Repeated	0.01810 ± 0.00000	0.00359 ± 0.00000	0.47968 ± 0.00185
Reversal	Reset	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.65722 ± 0.00820
Reversal	Reset	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.71237 ± 0.00297
Reversal	Reset	Neyman	0.01830 ± 0.00000	0.00000 ± 0.00000	0.71102 ± 0.00318
Reversal	Reset	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.71445 ± 0.00250
Reversal	Reset	Once	0.01827 ± 0.00001	0.00060 ± 0.00000	0.69498 ± 0.00676
Reversal	Reset	Repeated	0.01811 ± 0.00000	0.00361 ± 0.00000	0.69142 ± 0.00633
Reversal	Cumulative	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.49808 ± 0.00972
Reversal	Cumulative	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.48280 ± 0.00189
Reversal	Cumulative	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.48226 ± 0.00092
Reversal	Cumulative	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.49484 ± 0.00631
Reversal	Cumulative	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.48384 ± 0.00307
Reversal	Cumulative	Repeated	0.01809 ± 0.00000	0.00359 ± 0.00000	0.49785 ± 0.00945
Reversal	Window 8	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72062 ± 0.00152
Reversal	Window 8	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.72163 ± 0.00000
Reversal	Window 8	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Reversal	Window 8	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.72195 ± 0.00000
Reversal	Window 8	Once	0.01824 ± 0.00001	0.00060 ± 0.00000	0.72056 ± 0.00151
Reversal	Window 8	Repeated	0.01810 ± 0.00000	0.00359 ± 0.00000	0.72190 ± 0.00000
Uniform gain	Reset	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.65722 ± 0.00820

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Environment	Memory	Arm (SR = Successive rejection)	Expense	Trial expense	Late net
Uniform gain	Reset	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.71237 ± 0.00297
Uniform gain	Reset	Neyman	0.01831 ± 0.00000	0.00000 ± 0.00000	0.71102 ± 0.00318
Uniform gain	Reset	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.71445 ± 0.00250
Uniform gain	Reset	Once	0.01826 ± 0.00001	0.00060 ± 0.00000	0.69077 ± 0.00763
Uniform gain	Reset	Repeated	0.01811 ± 0.00000	0.00361 ± 0.00000	0.69142 ± 0.00633
Uniform gain	Cumulative	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Uniform gain	Cumulative	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.72163 ± 0.00000
Uniform gain	Cumulative	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Uniform gain	Cumulative	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.72195 ± 0.00000
Uniform gain	Cumulative	Once	0.01823 ± 0.00001	0.00060 ± 0.00000	0.72175 ± 0.00001
Uniform gain	Cumulative	Repeated	0.01809 ± 0.00000	0.00359 ± 0.00000	0.72191 ± 0.00000
Uniform gain	Window 8	Biased	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72062 ± 0.00152
Uniform gain	Window 8	Balanced	0.01837 ± 0.00000	0.00000 ± 0.00000	0.72163 ± 0.00000
Uniform gain	Window 8	Neyman	0.01821 ± 0.00000	0.00000 ± 0.00000	0.72179 ± 0.00000
Uniform gain	Window 8	SR	0.01805 ± 0.00000	0.00000 ± 0.00000	0.72195 ± 0.00000
Uniform gain	Window 8	Once	0.01823 ± 0.00001	0.00060 ± 0.00000	0.72175 ± 0.00001
Uniform gain	Window 8	Repeated	0.01810 ± 0.00000	0.00359 ± 0.00000	0.72190 ± 0.00000

Reversal labels in Panel C give the one-based round before which the change occurs.

C. Paired net-value contrasts

Study	Environment	Memory	Comparison	N	Difference
core	Stationary	Reset	Repeated - Balanced	128	-0.01702 ± 0.00262
core	Stationary	Reset	Repeated - Successive rejection	128	-0.01878 ± 0.00266
core	Stationary	Reset	Repeated - Discover once	128	-0.00142 ± 0.00441
core	Stationary	Reset	Repeated - uniform fixed mixture	128	0.00096 ± 0.00252
core	Stationary	Cumulative	Repeated - Balanced	128	-0.00007 ± 0.00092
core	Stationary	Cumulative	Repeated - Successive rejection	128	-0.00059 ± 0.00097
core	Stationary	Cumulative	Repeated - Discover once	128	0.00071 ± 0.00070
core	Stationary	Cumulative	Repeated - uniform fixed mixture	128	0.00343 ± 0.00117
core	Stationary	Window 8	Repeated - Balanced	128	-0.00130 ± 0.00114
core	Stationary	Window 8	Repeated - Successive rejection	128	-0.00212 ± 0.00115
core	Stationary	Window 8	Repeated - Discover once	128	0.00254 ± 0.00166
core	Stationary	Window 8	Repeated - uniform fixed mixture	128	0.00518 ± 0.00137
core	Stationary	Cumulative	Balanced - Reset	128	0.02783 ± 0.00124
core	Stationary	Window 8	Balanced - Reset	128	0.02697 ± 0.00126
core	Stationary	Cumulative	Repeated discovery - Reset	128	0.04478 ± 0.00283
core	Stationary	Window 8	Repeated discovery - Reset	128	0.04268 ± 0.00291
core	Stationary	Cumulative	Discover once - Reset	128	0.04264 ± 0.00456
core	Stationary	Window 8	Discover once - Reset	128	0.03873 ± 0.00364
core	Reversal at 25	Reset	Repeated - Balanced	128	-0.01997 ± 0.00235
core	Reversal at 25	Reset	Repeated - Successive rejection	128	-0.02153 ± 0.00241
core	Reversal at 25	Reset	Repeated - Discover once	128	-0.00159 ± 0.00500
core	Reversal at 25	Reset	Repeated - uniform fixed mixture	128	-0.00109 ± 0.00224
core	Reversal at 25	Cumulative	Repeated - Balanced	128	0.00321 ± 0.00245
core	Reversal at 25	Cumulative	Repeated - Successive rejection	128	0.00062 ± 0.00273
core	Reversal at 25	Cumulative	Repeated - Discover once	128	0.00384 ± 0.00250
core	Reversal at 25	Cumulative	Repeated - uniform fixed mixture	128	0.00583 ± 0.00259
core	Reversal at 25	Window 8	Repeated - Balanced	128	0.00623 ± 0.00240
core	Reversal at 25	Window 8	Repeated - Successive rejection	128	0.00347 ± 0.00252
core	Reversal at 25	Window 8	Repeated - Discover once	128	0.00975 ± 0.00267
core	Reversal at 25	Window 8	Repeated - uniform fixed mixture	128	0.01114 ± 0.00255
core	Reversal at 25	Cumulative	Balanced - Reset	128	-0.10253 ± 0.00132
core	Reversal at 25	Window 8	Balanced - Reset	128	-0.00697 ± 0.00136
core	Reversal at 25	Cumulative	Repeated discovery - Reset	128	-0.07935 ± 0.00345
core	Reversal at 25	Window 8	Repeated discovery - Reset	128	0.01924 ± 0.00351
core	Reversal at 25	Cumulative	Discover once - Reset	128	-0.08478 ± 0.00493

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Study	Environment	Memory	Comparison	N	Difference
core	Reversal at 25	Window 8	Discover once - Reset	128	0.00789 ± 0.00423
core	Uniform gain	Reset	Repeated - Balanced	128	-0.02000 ± 0.00240
core	Uniform gain	Reset	Repeated - Successive rejection	128	-0.02208 ± 0.00239
core	Uniform gain	Reset	Repeated - Discover once	128	-0.00204 ± 0.00464
core	Uniform gain	Reset	Repeated - uniform fixed mixture	128	-0.00220 ± 0.00223
core	Uniform gain	Cumulative	Repeated - Balanced	128	0.00039 ± 0.00013
core	Uniform gain	Cumulative	Repeated - Successive rejection	128	0.00003 ± 0.00011
core	Uniform gain	Cumulative	Repeated - Discover once	128	0.00013 ± 0.00001
core	Uniform gain	Cumulative	Repeated - uniform fixed mixture	128	0.00139 ± 0.00044
core	Uniform gain	Window 8	Repeated - Balanced	128	0.00039 ± 0.00013
core	Uniform gain	Window 8	Repeated - Successive rejection	128	0.00003 ± 0.00011
core	Uniform gain	Window 8	Repeated - Discover once	128	0.00085 ± 0.00080
core	Uniform gain	Window 8	Repeated - uniform fixed mixture	128	0.00197 ± 0.00056
core	Uniform gain	Cumulative	Balanced - Reset	128	0.00736 ± 0.00110
core	Uniform gain	Window 8	Balanced - Reset	128	0.00736 ± 0.00110
core	Uniform gain	Cumulative	Repeated discovery - Reset	128	0.02775 ± 0.00235
core	Uniform gain	Window 8	Repeated discovery - Reset	128	0.02775 ± 0.00235
core	Uniform gain	Cumulative	Discover once - Reset	128	0.02558 ± 0.00459
core	Uniform gain	Window 8	Discover once - Reset	128	0.02486 ± 0.00437
exploratory control	Stationary	Reset	Repeated - Balanced	128	-0.01665 ± 0.00255
exploratory control	Stationary	Reset	Repeated - Trial-matched Balanced	128	-0.01382 ± 0.00238
exploratory control	Stationary	Reset	Repeated - Discover once	128	0.00035 ± 0.00448
exploratory control	Stationary	Cumulative	Repeated - Balanced	128	0.00078 ± 0.00091
exploratory control	Stationary	Cumulative	Repeated - Trial-matched Balanced	128	-0.00016 ± 0.00046
exploratory control	Stationary	Cumulative	Repeated - Discover once	128	0.00110 ± 0.00135
exploratory control	Stationary	Window 8	Repeated - Balanced	128	0.00023 ± 0.00104
exploratory control	Stationary	Window 8	Repeated - Trial-matched Balanced	128	-0.00116 ± 0.00058
exploratory control	Stationary	Window 8	Repeated - Discover once	128	0.00538 ± 0.00210
exploratory control	Reversal at 25	Reset	Repeated - Balanced	128	-0.01808 ± 0.00239
exploratory control	Reversal at 25	Reset	Repeated - Trial-matched Balanced	128	-0.01525 ± 0.00224
exploratory control	Reversal at 25	Reset	Repeated - Discover once	128	-0.00223 ± 0.00438
exploratory control	Reversal at 25	Cumulative	Repeated - Balanced	128	0.00922 ± 0.00353
exploratory control	Reversal at 25	Cumulative	Repeated - Trial-matched Balanced	128	0.00832 ± 0.00363
exploratory control	Reversal at 25	Cumulative	Repeated - Discover once	128	0.00926 ± 0.00397
exploratory control	Reversal at 25	Window 8	Repeated - Balanced	128	0.00607 ± 0.00217
exploratory control	Reversal at 25	Window 8	Repeated - Trial-matched Balanced	128	0.00191 ± 0.00207
exploratory control	Reversal at 25	Window 8	Repeated - Discover once	128	0.00889 ± 0.00220
exploratory control	Reversal at 21	Reset	Repeated - Balanced	128	-0.01817 ± 0.00254
exploratory control	Reversal at 21	Reset	Repeated - Trial-matched Balanced	128	-0.01586 ± 0.00228
exploratory control	Reversal at 21	Reset	Repeated - Discover once	128	-0.00242 ± 0.00439
exploratory control	Reversal at 21	Cumulative	Repeated - Balanced	128	0.00840 ± 0.00513
exploratory control	Reversal at 21	Cumulative	Repeated - Trial-matched Balanced	128	0.01132 ± 0.00526
exploratory control	Reversal at 21	Cumulative	Repeated - Discover once	128	0.00891 ± 0.00550
exploratory control	Reversal at 21	Window 8	Repeated - Balanced	128	0.00031 ± 0.00118
exploratory control	Reversal at 21	Window 8	Repeated - Trial-matched Balanced	128	-0.00063 ± 0.00072
exploratory control	Reversal at 21	Window 8	Repeated - Discover once	128	0.00257 ± 0.00137
exploratory control	Reversal at 29	Reset	Repeated - Balanced	128	-0.01700 ± 0.00252
exploratory control	Reversal at 29	Reset	Repeated - Trial-matched Balanced	128	-0.01399 ± 0.00235
exploratory control	Reversal at 29	Reset	Repeated - Discover once	128	-0.00116 ± 0.00433
exploratory control	Reversal at 29	Cumulative	Repeated - Balanced	128	0.00262 ± 0.00148
exploratory control	Reversal at 29	Cumulative	Repeated - Trial-matched Balanced	128	0.00168 ± 0.00136
exploratory control	Reversal at 29	Cumulative	Repeated - Discover once	128	0.00293 ± 0.00225
exploratory control	Reversal at 29	Window 8	Repeated - Balanced	128	0.00004 ± 0.00120
exploratory control	Reversal at 29	Window 8	Repeated - Trial-matched Balanced	128	-0.00057 ± 0.00069
exploratory control	Reversal at 29	Window 8	Repeated - Discover once	128	0.00342 ± 0.00177

Table S2: Program selection at the final scheduled assessment

Source: `results/learning_conditional.csv`.

Each row summarizes 128 runs at the final program assessment: round 1 for discover-once and round 41 for repeated discovery. Program columns give the percentage retaining Biased, Balanced, or Neyman. The source table also records the conditional block outcomes and earlier assessments.

Final program-selection proportions

Environment	Memory	Arm	Review	Biased (%)	Balanced (%)	Neyman (%)
Stationary	Reset	Once	1	27.3	37.5	35.2
Stationary	Reset	Repeated	41	27.3	43.0	29.7
Stationary	Cumulative	Once	1	27.3	37.5	35.2
Stationary	Cumulative	Repeated	41	38.3	31.2	30.5
Stationary	Window 8	Once	1	27.3	37.5	35.2
Stationary	Window 8	Repeated	41	32.0	29.7	38.3
Reversal	Reset	Once	1	27.3	37.5	35.2
Reversal	Reset	Repeated	41	47.7	22.7	29.7
Reversal	Cumulative	Once	1	27.3	37.5	35.2
Reversal	Cumulative	Repeated	41	38.3	31.2	30.5
Reversal	Window 8	Once	1	27.3	37.5	35.2
Reversal	Window 8	Repeated	41	47.7	20.3	32.0
Uniform gain	Reset	Once	1	35.9	27.3	36.7
Uniform gain	Reset	Repeated	41	47.7	22.7	29.7
Uniform gain	Cumulative	Once	1	35.9	27.3	36.7
Uniform gain	Cumulative	Repeated	41	30.5	25.8	43.8
Uniform gain	Window 8	Once	1	35.9	27.3	36.7
Uniform gain	Window 8	Repeated	41	47.7	20.3	32.0

Table S3: Audit policy and memory-window grid

Source: `results/robustness_summary.csv`.

All 54 combinations of three environments, three memory windows, and six audit policies. Each cell uses 200 independent seeds. Paired differences compare with fixed audit rate .10 within the same environment and window. The outcome is net value per task under the supplementary audit model.

Complete audit grid

Environment	Window	Policy	Net value	Difference vs. fixed .10
Stationary	6	No audit	0.88054 ± 0.00117	0.02452 ± 0.00080
Stationary	6	Fixed 0.05	0.85918 ± 0.00181	0.00316 ± 0.00119
Stationary	6	Fixed 0.10	0.85602 ± 0.00152	0.00000 ± 0.00000
Stationary	6	Fixed 0.40	0.80061 ± 0.00120	-0.05541 ± 0.00078
Stationary	6	Periodic dense	0.84527 ± 0.00148	-0.01075 ± 0.00108
Stationary	6	Adaptive audit	0.85356 ± 0.00135	-0.00246 ± 0.00082
Stationary	12	No audit	0.88054 ± 0.00117	0.02118 ± 0.00047
Stationary	12	Fixed 0.05	0.86517 ± 0.00177	0.00580 ± 0.00109
Stationary	12	Fixed 0.10	0.85937 ± 0.00130	0.00000 ± 0.00000
Stationary	12	Fixed 0.40	0.80066 ± 0.00119	-0.05871 ± 0.00049
Stationary	12	Periodic dense	0.85290 ± 0.00120	-0.00647 ± 0.00051
Stationary	12	Adaptive audit	0.86681 ± 0.00127	0.00744 ± 0.00060
Stationary	24	No audit	0.88054 ± 0.00117	0.02037 ± 0.00027
Stationary	24	Fixed 0.05	0.86869 ± 0.00147	0.00851 ± 0.00067
Stationary	24	Fixed 0.10	0.86018 ± 0.00120	0.00000 ± 0.00000
Stationary	24	Fixed 0.40	0.80066 ± 0.00119	-0.05952 ± 0.00031
Stationary	24	Periodic dense	0.85319 ± 0.00118	-0.00698 ± 0.00028
Stationary	24	Adaptive audit	0.87162 ± 0.00125	0.01144 ± 0.00037
Slow shifts	6	No audit	0.56603 ± 0.00195	-0.09302 ± 0.00284
Slow shifts	6	Fixed 0.05	0.64384 ± 0.00319	-0.01521 ± 0.00231
Slow shifts	6	Fixed 0.10	0.65905 ± 0.00249	0.00000 ± 0.00000
Slow shifts	6	Fixed 0.40	0.62772 ± 0.00153	-0.03133 ± 0.00222
Slow shifts	6	Periodic dense	0.64667 ± 0.00239	-0.01238 ± 0.00245
Slow shifts	6	Adaptive audit	0.66438 ± 0.00198	0.00533 ± 0.00234
Slow shifts	12	No audit	0.56603 ± 0.00195	-0.09364 ± 0.00277
Slow shifts	12	Fixed 0.05	0.65240 ± 0.00322	-0.00727 ± 0.00266
Slow shifts	12	Fixed 0.10	0.65967 ± 0.00241	0.00000 ± 0.00000
Slow shifts	12	Fixed 0.40	0.61216 ± 0.00148	-0.04751 ± 0.00202
Slow shifts	12	Periodic dense	0.66029 ± 0.00204	0.00062 ± 0.00236
Slow shifts	12	Adaptive audit	0.66785 ± 0.00239	0.00818 ± 0.00263
Slow shifts	24	No audit	0.56603 ± 0.00195	-0.06813 ± 0.00271
Slow shifts	24	Fixed 0.05	0.63665 ± 0.00326	0.00249 ± 0.00266
Slow shifts	24	Fixed 0.10	0.63416 ± 0.00242	0.00000 ± 0.00000
Slow shifts	24	Fixed 0.40	0.57706 ± 0.00157	-0.05710 ± 0.00205
Slow shifts	24	Periodic dense	0.63149 ± 0.00278	-0.00267 ± 0.00300
Slow shifts	24	Adaptive audit	0.65398 ± 0.00290	0.01982 ± 0.00316
Fast shifts	6	No audit	0.56512 ± 0.00210	-0.02974 ± 0.00259
Fast shifts	6	Fixed 0.05	0.59498 ± 0.00279	0.00012 ± 0.00226
Fast shifts	6	Fixed 0.10	0.59486 ± 0.00251	0.00000 ± 0.00000
Fast shifts	6	Fixed 0.40	0.54718 ± 0.00180	-0.04768 ± 0.00209
Fast shifts	6	Periodic dense	0.58839 ± 0.00256	-0.00647 ± 0.00247
Fast shifts	6	Adaptive audit	0.59544 ± 0.00234	0.00058 ± 0.00247
Fast shifts	12	No audit	0.56512 ± 0.00210	0.03871 ± 0.00253
Fast shifts	12	Fixed 0.05	0.54272 ± 0.00307	0.01632 ± 0.00243
Fast shifts	12	Fixed 0.10	0.52640 ± 0.00295	0.00000 ± 0.00000
Fast shifts	12	Fixed 0.40	0.45976 ± 0.00214	-0.06665 ± 0.00221
Fast shifts	12	Periodic dense	0.52383 ± 0.00234	-0.00257 ± 0.00265
Fast shifts	12	Adaptive audit	0.52467 ± 0.00273	-0.00173 ± 0.00289
Fast shifts	24	No audit	0.56512 ± 0.00210	-0.01550 ± 0.00239
Fast shifts	24	Fixed 0.05	0.58156 ± 0.00266	0.00093 ± 0.00242
Fast shifts	24	Fixed 0.10	0.58062 ± 0.00237	0.00000 ± 0.00000
Fast shifts	24	Fixed 0.40	0.54112 ± 0.00239	-0.03950 ± 0.00270
Fast shifts	24	Periodic dense	0.57131 ± 0.00240	-0.00931 ± 0.00244
Fast shifts	24	Adaptive audit	0.56229 ± 0.00220	-0.01834 ± 0.00251

Table S4: Expected-value-of-information sensitivity

Source: `results/audit_extensions_summary.csv`.

All 72 combinations of three priors, two routing thresholds, two candidate-generation costs, two environments, and three policies. Each cell uses 96 independent test seeds; development seeds selected the horizon separately. Nonzero generation cost charges counterfactual A outputs when route B is executed. Paired differences use fixed audit rate .10 within the same condition.

Complete prior, threshold, and generation-cost grid

Prior	Threshold	g_c	Environment	Policy	Net value	Difference vs. fixed .10
1:1	0.14	0	Stationary	Fixed 0.10	0.84794 ± 0.00236	0.00000 ± 0.00000
1:1	0.14	0	Stationary	Adaptive audit	0.85300 ± 0.00213	0.00506 ± 0.00173
1:1	0.14	0	Stationary	EVSI	0.83379 ± 0.00359	-0.01415 ± 0.00326
1:1	0.14	0	Slow shifts	Fixed 0.10	0.66159 ± 0.00308	0.00000 ± 0.00000
1:1	0.14	0	Slow shifts	Adaptive audit	0.67534 ± 0.00293	0.01375 ± 0.00317
1:1	0.14	0	Slow shifts	EVSI	0.67584 ± 0.00569	0.01426 ± 0.00540
1:1	0.14	0.2	Stationary	Fixed 0.10	0.84718 ± 0.00244	0.00000 ± 0.00000
1:1	0.14	0.2	Stationary	Adaptive audit	0.85173 ± 0.00220	0.00455 ± 0.00184
1:1	0.14	0.2	Stationary	EVSI	0.82623 ± 0.00362	-0.02094 ± 0.00338
1:1	0.14	0.2	Slow shifts	Fixed 0.10	0.65182 ± 0.00305	0.00000 ± 0.00000
1:1	0.14	0.2	Slow shifts	Adaptive audit	0.66587 ± 0.00299	0.01405 ± 0.00323
1:1	0.14	0.2	Slow shifts	EVSI	0.66778 ± 0.00568	0.01596 ± 0.00542
1:1	0.16	0	Stationary	Fixed 0.10	0.85228 ± 0.00191	0.00000 ± 0.00000
1:1	0.16	0	Stationary	Adaptive audit	0.85706 ± 0.00194	0.00479 ± 0.00134
1:1	0.16	0	Stationary	EVSI	0.83421 ± 0.00355	-0.01807 ± 0.00313
1:1	0.16	0	Slow shifts	Fixed 0.10	0.65968 ± 0.00334	0.00000 ± 0.00000
1:1	0.16	0	Slow shifts	Adaptive audit	0.67286 ± 0.00277	0.01318 ± 0.00371
1:1	0.16	0	Slow shifts	EVSI	0.67570 ± 0.00566	0.01602 ± 0.00547
1:1	0.16	0.2	Stationary	Fixed 0.10	0.85180 ± 0.00195	0.00000 ± 0.00000
1:1	0.16	0.2	Stationary	Adaptive audit	0.85607 ± 0.00199	0.00427 ± 0.00145
1:1	0.16	0.2	Stationary	EVSI	0.82674 ± 0.00357	-0.02506 ± 0.00321
1:1	0.16	0.2	Slow shifts	Fixed 0.10	0.65064 ± 0.00327	0.00000 ± 0.00000
1:1	0.16	0.2	Slow shifts	Adaptive audit	0.66208 ± 0.00286	0.01143 ± 0.00367
1:1	0.16	0.2	Slow shifts	EVSI	0.66745 ± 0.00563	0.01681 ± 0.00541
1:9	0.14	0	Stationary	Fixed 0.10	0.85814 ± 0.00172	0.00000 ± 0.00000
1:9	0.14	0	Stationary	Adaptive audit	0.86494 ± 0.00192	0.00680 ± 0.00087
1:9	0.14	0	Stationary	EVSI	0.86557 ± 0.00207	0.00743 ± 0.00147
1:9	0.14	0	Slow shifts	Fixed 0.10	0.66200 ± 0.00333	0.00000 ± 0.00000
1:9	0.14	0	Slow shifts	Adaptive audit	0.67056 ± 0.00285	0.00856 ± 0.00352
1:9	0.14	0	Slow shifts	EVSI	0.65987 ± 0.00606	-0.00213 ± 0.00667
1:9	0.14	0.2	Stationary	Fixed 0.10	0.85805 ± 0.00173	0.00000 ± 0.00000
1:9	0.14	0.2	Stationary	Adaptive audit	0.86480 ± 0.00195	0.00675 ± 0.00094
1:9	0.14	0.2	Stationary	EVSI	0.86507 ± 0.00222	0.00702 ± 0.00164
1:9	0.14	0.2	Slow shifts	Fixed 0.10	0.65364 ± 0.00323	0.00000 ± 0.00000
1:9	0.14	0.2	Slow shifts	Adaptive audit	0.65984 ± 0.00290	0.00619 ± 0.00345
1:9	0.14	0.2	Slow shifts	EVSI	0.65465 ± 0.00638	0.00100 ± 0.00689
1:9	0.16	0	Stationary	Fixed 0.10	0.85918 ± 0.00162	0.00000 ± 0.00000
1:9	0.16	0	Stationary	Adaptive audit	0.86861 ± 0.00183	0.00943 ± 0.00053
1:9	0.16	0	Stationary	EVSI	0.86603 ± 0.00210	0.00685 ± 0.00134
1:9	0.16	0	Slow shifts	Fixed 0.10	0.65150 ± 0.00390	0.00000 ± 0.00000
1:9	0.16	0	Slow shifts	Adaptive audit	0.66416 ± 0.00362	0.01266 ± 0.00468
1:9	0.16	0	Slow shifts	EVSI	0.65263 ± 0.00621	0.00114 ± 0.00705
1:9	0.16	0.2	Stationary	Fixed 0.10	0.85915 ± 0.00162	0.00000 ± 0.00000
1:9	0.16	0.2	Stationary	Adaptive audit	0.86858 ± 0.00184	0.00943 ± 0.00055
1:9	0.16	0.2	Stationary	EVSI	0.86603 ± 0.00210	0.00687 ± 0.00135
1:9	0.16	0.2	Slow shifts	Fixed 0.10	0.64420 ± 0.00377	0.00000 ± 0.00000

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Prior	Threshold	g_c	Environment	Policy	Net value	Difference vs. fixed .10
1:9	0.16	0.2	Slow shifts	Adaptive audit	0.65145 ± 0.00364	0.00725 ± 0.00449
1:9	0.16	0.2	Slow shifts	EVSI	0.65263 ± 0.00621	0.00844 ± 0.00697
1:19	0.14	0	Stationary	Fixed 0.10	0.85951 ± 0.00162	0.00000 ± 0.00000
1:19	0.14	0	Stationary	Adaptive audit	0.87199 ± 0.00177	0.01248 ± 0.00036
1:19	0.14	0	Stationary	EVSI	0.87950 ± 0.00163	0.02000 ± 0.00020
1:19	0.14	0	Slow shifts	Fixed 0.10	0.64678 ± 0.00420	0.00000 ± 0.00000
1:19	0.14	0	Slow shifts	Adaptive audit	0.65715 ± 0.00397	0.01037 ± 0.00451
1:19	0.14	0	Slow shifts	EVSI	0.56729 ± 0.00321	-0.07949 ± 0.00502
1:19	0.14	0.2	Stationary	Fixed 0.10	0.85951 ± 0.00162	0.00000 ± 0.00000
1:19	0.14	0.2	Stationary	Adaptive audit	0.87198 ± 0.00177	0.01247 ± 0.00037
1:19	0.14	0.2	Stationary	EVSI	0.87950 ± 0.00163	0.02000 ± 0.00020
1:19	0.14	0.2	Slow shifts	Fixed 0.10	0.63993 ± 0.00406	0.00000 ± 0.00000
1:19	0.14	0.2	Slow shifts	Adaptive audit	0.64531 ± 0.00381	0.00539 ± 0.00416
1:19	0.14	0.2	Slow shifts	EVSI	0.56729 ± 0.00321	-0.07263 ± 0.00487
1:19	0.16	0	Stationary	Fixed 0.10	0.85954 ± 0.00163	0.00000 ± 0.00000
1:19	0.16	0	Stationary	Adaptive audit	0.87348 ± 0.00167	0.01394 ± 0.00026
1:19	0.16	0	Stationary	EVSI	0.87950 ± 0.00163	0.01997 ± 0.00019
1:19	0.16	0	Slow shifts	Fixed 0.10	0.62630 ± 0.00474	0.00000 ± 0.00000
1:19	0.16	0	Slow shifts	Adaptive audit	0.63964 ± 0.00445	0.01334 ± 0.00580
1:19	0.16	0	Slow shifts	EVSI	0.56729 ± 0.00321	-0.05901 ± 0.00535
1:19	0.16	0.2	Stationary	Fixed 0.10	0.85954 ± 0.00163	0.00000 ± 0.00000
1:19	0.16	0.2	Stationary	Adaptive audit	0.87348 ± 0.00167	0.01394 ± 0.00026
1:19	0.16	0.2	Stationary	EVSI	0.87950 ± 0.00163	0.01997 ± 0.00019
1:19	0.16	0.2	Slow shifts	Fixed 0.10	0.62102 ± 0.00454	0.00000 ± 0.00000
1:19	0.16	0.2	Slow shifts	Adaptive audit	0.62680 ± 0.00414	0.00578 ± 0.00529
1:19	0.16	0.2	Slow shifts	EVSI	0.56729 ± 0.00321	-0.05373 ± 0.00515

Table S5: Study 2: review arrangements and evidence generation

Source: `study2/results/learning_summary.csv`; `study2/results/paired_contrasts.csv`.

All 84 environment–policy–memory conditions: six environments, four paid policies crossed with three memories, and two fixed baselines per environment. Each learning condition uses 128 trajectories of 48 rounds; late outcomes cover the final 16 rounds. Error and optimal-choice hit are percentages. Regret is deployment loss above the finite-catalog minimum, excluding acquisition and switching. Acquisition cost is per round; production cost is per task in human-call units. Fixed baselines use exact expected-outcome scores. Paired, Randomized, Omit echo and Final only denote the four observation policies defined in Appendix E.

A. Full-horizon selection and net value

Environment	Policy	Memory	Error (%)	Net value	Hit (%)
No echo	Final only	Reset	10.472 ± 0.056	0.62353 ± 0.00174	0.03 ± 0.04
No echo	Final only	Cumulative	4.919 ± 0.197	0.78045 ± 0.00563	14.47 ± 3.55
No echo	Final only	Window 8	7.116 ± 0.137	0.71668 ± 0.00390	6.38 ± 0.76
No echo	Paired	Reset	1.864 ± 0.018	0.85928 ± 0.00056	33.06 ± 1.16
No echo	Paired	Cumulative	1.772 ± 0.002	0.86475 ± 0.00008	98.70 ± 0.18
No echo	Paired	Window 8	1.772 ± 0.002	0.86475 ± 0.00008	98.70 ± 0.18
No echo	Randomized	Reset	1.944 ± 0.022	0.85786 ± 0.00068	50.85 ± 1.21
No echo	Randomized	Cumulative	1.772 ± 0.003	0.86333 ± 0.00026	56.67 ± 6.18
No echo	Randomized	Window 8	1.772 ± 0.003	0.86321 ± 0.00016	53.99 ± 3.14
No echo	Omit echo	Reset	1.857 ± 0.018	0.86217 ± 0.00056	98.36 ± 0.33
No echo	Omit echo	Cumulative	1.772 ± 0.002	0.86480 ± 0.00008	99.97 ± 0.04
No echo	Omit echo	Window 8	1.772 ± 0.002	0.86480 ± 0.00008	99.97 ± 0.04
No echo	Fixed exposed	Fixed	3.520 ± 0.000	0.87440 ± 0.00000	0.00 ± 0.00
No echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
High echo	Final only	Reset	11.109 ± 0.058	0.60564 ± 0.00184	0.00 ± 0.00
High echo	Final only	Cumulative	5.983 ± 0.217	0.74760 ± 0.00640	10.89 ± 3.09
High echo	Final only	Window 8	8.788 ± 0.124	0.66901 ± 0.00365	2.70 ± 0.50
High echo	Paired	Reset	1.880 ± 0.017	0.85753 ± 0.00052	97.92 ± 0.31
High echo	Paired	Cumulative	1.773 ± 0.003	0.86077 ± 0.00009	99.95 ± 0.05
High echo	Paired	Window 8	1.773 ± 0.003	0.86077 ± 0.00009	99.95 ± 0.05
High echo	Randomized	Reset	2.267 ± 0.045	0.84636 ± 0.00134	92.53 ± 0.64
High echo	Randomized	Cumulative	1.782 ± 0.008	0.86078 ± 0.00024	99.82 ± 0.12
High echo	Randomized	Window 8	1.782 ± 0.008	0.86078 ± 0.00024	99.82 ± 0.12
High echo	Omit echo	Reset	9.961 ± 0.023	0.61912 ± 0.00070	0.00 ± 0.00
High echo	Omit echo	Cumulative	9.825 ± 0.004	0.62323 ± 0.00013	0.00 ± 0.00
High echo	Omit echo	Window 8	9.825 ± 0.004	0.62323 ± 0.00013	0.00 ± 0.00
High echo	Fixed exposed	Fixed	9.637 ± 0.000	0.69089 ± 0.00000	0.00 ± 0.00
High echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Both dependences	Final only	Reset	11.045 ± 0.054	0.60834 ± 0.00170	0.00 ± 0.00
Both dependences	Final only	Cumulative	9.728 ± 0.136	0.63630 ± 0.00404	5.49 ± 2.55
Both dependences	Final only	Window 8	10.295 ± 0.079	0.62548 ± 0.00251	1.01 ± 0.27
Both dependences	Paired	Reset	7.822 ± 0.017	0.68223 ± 0.00036	51.84 ± 1.28
Both dependences	Paired	Cumulative	7.423 ± 0.049	0.69099 ± 0.00105	82.29 ± 3.92
Both dependences	Paired	Window 8	7.563 ± 0.034	0.68789 ± 0.00073	70.72 ± 2.72
Both dependences	Randomized	Reset	8.265 ± 0.041	0.66972 ± 0.00115	46.96 ± 1.21
Both dependences	Randomized	Cumulative	7.498 ± 0.063	0.68945 ± 0.00135	77.10 ± 4.85
Both dependences	Randomized	Window 8	7.631 ± 0.040	0.68650 ± 0.00087	66.28 ± 3.03
Both dependences	Omit echo	Reset	13.749 ± 0.098	0.50789 ± 0.00302	0.00 ± 0.00
Both dependences	Omit echo	Cumulative	15.939 ± 0.317	0.43965 ± 0.00969	0.00 ± 0.00
Both dependences	Omit echo	Window 8	15.125 ± 0.212	0.46465 ± 0.00649	0.00 ± 0.00
Both dependences	Fixed exposed	Fixed	9.919 ± 0.000	0.68242 ± 0.00000	0.00 ± 0.00
Both dependences	Fixed blind	Fixed	8.560 ± 0.000	0.72020 ± 0.00000	0.00 ± 0.00
Echo shift	Final only	Reset	10.722 ± 0.054	0.61665 ± 0.00170	0.00 ± 0.00
Echo shift	Final only	Cumulative	5.648 ± 0.217	0.75770 ± 0.00621	11.20 ± 2.78
Echo shift	Final only	Window 8	7.826 ± 0.118	0.69639 ± 0.00341	4.82 ± 0.57

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Environment	Policy	Memory	Error (%)	Net value	Hit (%)
Echo shift	Paired	Reset	1.886 ± 0.020	0.85798 ± 0.00060	65.32 ± 0.88
Echo shift	Paired	Cumulative	1.779 ± 0.006	0.86255 ± 0.00018	98.54 ± 0.18
Echo shift	Paired	Window 8	1.772 ± 0.002	0.86274 ± 0.00008	98.62 ± 0.17
Echo shift	Randomized	Reset	2.148 ± 0.036	0.85081 ± 0.00108	71.22 ± 0.95
Echo shift	Randomized	Cumulative	2.113 ± 0.083	0.85206 ± 0.00238	72.40 ± 2.66
Echo shift	Randomized	Window 8	1.960 ± 0.040	0.85659 ± 0.00117	74.54 ± 2.11
Echo shift	Omit echo	Reset	5.926 ± 0.023	0.74011 ± 0.00071	48.96 ± 0.26
Echo shift	Omit echo	Cumulative	5.797 ± 0.002	0.74404 ± 0.00008	49.97 ± 0.04
Echo shift	Omit echo	Window 8	5.797 ± 0.002	0.74404 ± 0.00008	49.97 ± 0.04
Echo shift	Fixed exposed	Fixed	6.579 ± 0.000	0.78264 ± 0.00000	0.00 ± 0.00
Echo shift	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Source asymmetry	Final only	Reset	11.009 ± 0.058	0.60849 ± 0.00181	0.00 ± 0.00
Source asymmetry	Final only	Cumulative	6.352 ± 0.191	0.73785 ± 0.00535	5.34 ± 2.06
Source asymmetry	Final only	Window 8	8.247 ± 0.134	0.68513 ± 0.00385	1.84 ± 0.41
Source asymmetry	Paired	Reset	2.983 ± 0.028	0.82450 ± 0.00084	90.67 ± 0.77
Source asymmetry	Paired	Cumulative	2.693 ± 0.004	0.83318 ± 0.00013	99.76 ± 0.13
Source asymmetry	Paired	Window 8	2.693 ± 0.004	0.83318 ± 0.00013	99.76 ± 0.13
Source asymmetry	Randomized	Reset	3.414 ± 0.035	0.81334 ± 0.00107	48.91 ± 1.38
Source asymmetry	Randomized	Cumulative	2.936 ± 0.046	0.82728 ± 0.00116	70.91 ± 5.42
Source asymmetry	Randomized	Window 8	3.009 ± 0.027	0.82545 ± 0.00069	62.27 ± 3.18
Source asymmetry	Omit echo	Reset	4.386 ± 0.094	0.78638 ± 0.00281	0.00 ± 0.00
Source asymmetry	Omit echo	Cumulative	3.546 ± 0.015	0.81160 ± 0.00044	0.00 ± 0.00
Source asymmetry	Omit echo	Window 8	3.546 ± 0.015	0.81160 ± 0.00044	0.00 ± 0.00
Source asymmetry	Fixed exposed	Fixed	5.101 ± 0.000	0.82697 ± 0.00000	0.00 ± 0.00
Source asymmetry	Fixed blind	Fixed	4.240 ± 0.000	0.84980 ± 0.00000	0.00 ± 0.00
Stronger humans	Final only	Reset	18.205 ± 0.073	0.39097 ± 0.00207	0.00 ± 0.00
Stronger humans	Final only	Cumulative	9.522 ± 0.263	0.63271 ± 0.00748	10.04 ± 3.28
Stronger humans	Final only	Window 8	12.956 ± 0.201	0.53738 ± 0.00577	3.79 ± 0.79
Stronger humans	Paired	Reset	5.413 ± 0.040	0.74875 ± 0.00121	77.65 ± 1.07
Stronger humans	Paired	Cumulative	4.623 ± 0.009	0.77306 ± 0.00028	99.37 ± 0.25
Stronger humans	Paired	Window 8	4.628 ± 0.011	0.77289 ± 0.00032	99.20 ± 0.29
Stronger humans	Randomized	Reset	5.629 ± 0.044	0.74241 ± 0.00133	72.31 ± 1.12
Stronger humans	Randomized	Cumulative	4.639 ± 0.014	0.77286 ± 0.00041	98.96 ± 0.35
Stronger humans	Randomized	Window 8	4.646 ± 0.014	0.77262 ± 0.00043	98.73 ± 0.38
Stronger humans	Omit echo	Reset	10.693 ± 0.101	0.59515 ± 0.00304	0.00 ± 0.00
Stronger humans	Omit echo	Cumulative	8.969 ± 0.024	0.64781 ± 0.00072	0.00 ± 0.00
Stronger humans	Omit echo	Window 8	8.982 ± 0.027	0.64741 ± 0.00082	0.00 ± 0.00
Stronger humans	Fixed exposed	Fixed	19.570 ± 0.000	0.39290 ± 0.00000	0.00 ± 0.00
Stronger humans	Fixed blind	Fixed	12.320 ± 0.000	0.60740 ± 0.00000	0.00 ± 0.00

B. Full-horizon costs and regret

Environment	Policy	Memory	Acquisition cost	Production cost	Regret
No echo	Final only	Reset	214.788 ± 0.076	0.4879 ± 0.0049	0.24580 ± 0.00174
No echo	Final only	Cumulative	214.788 ± 0.076	0.9764 ± 0.0341	0.08897 ± 0.00563
No echo	Final only	Window 8	214.788 ± 0.076	0.8658 ± 0.0193	0.15270 ± 0.00390
No echo	Paired	Reset	233.600 ± 0.000	1.3853 ± 0.0024	0.00552 ± 0.00056
No echo	Paired	Cumulative	233.600 ± 0.000	1.2526 ± 0.0004	0.00011 ± 0.00008
No echo	Paired	Window 8	233.600 ± 0.000	1.2526 ± 0.0004	0.00011 ± 0.00008
No echo	Randomized	Reset	232.500 ± 0.000	1.3501 ± 0.0028	0.00720 ± 0.00068
No echo	Randomized	Cumulative	232.500 ± 0.000	1.3367 ± 0.0124	0.00179 ± 0.00026
No echo	Randomized	Window 8	232.500 ± 0.000	1.3420 ± 0.0063	0.00189 ± 0.00016
No echo	Omit echo	Reset	233.600 ± 0.000	1.2538 ± 0.0010	0.00268 ± 0.00056
No echo	Omit echo	Cumulative	233.600 ± 0.000	1.2501 ± 0.0001	0.00005 ± 0.00008
No echo	Omit echo	Window 8	233.600 ± 0.000	1.2501 ± 0.0001	0.00005 ± 0.00008

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Environment	Policy	Memory	Acquisition cost	Production cost	Regret
No echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.04749 ± 0.00000
No echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.05049 ± 0.00000
High echo	Final only	Reset	209.381 ± 0.065	0.4931 ± 0.0055	0.26102 ± 0.00184
High echo	Final only	Cumulative	209.381 ± 0.065	1.0879 ± 0.0388	0.11915 ± 0.00640
High echo	Final only	Window 8	209.381 ± 0.065	0.8080 ± 0.0206	0.19770 ± 0.00365
High echo	Paired	Reset	233.600 ± 0.000	1.4516 ± 0.0006	0.00332 ± 0.00052
High echo	Paired	Cumulative	233.600 ± 0.000	1.4501 ± 0.0001	0.00008 ± 0.00009
High echo	Paired	Window 8	233.600 ± 0.000	1.4501 ± 0.0001	0.00008 ± 0.00009
High echo	Randomized	Reset	232.500 ± 0.000	1.4428 ± 0.0016	0.01475 ± 0.00134
High echo	Randomized	Cumulative	232.500 ± 0.000	1.4498 ± 0.0003	0.00035 ± 0.00024
High echo	Randomized	Window 8	232.500 ± 0.000	1.4498 ± 0.0003	0.00035 ± 0.00024
High echo	Omit echo	Reset	233.600 ± 0.000	1.2507 ± 0.0003	0.24173 ± 0.00070
High echo	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.23763 ± 0.00013
High echo	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.23763 ± 0.00013
High echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.22700 ± 0.00000
High echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04649 ± 0.00000
Both dependences	Final only	Reset	206.524 ± 0.051	0.4882 ± 0.0051	0.09392 ± 0.00170
Both dependences	Final only	Cumulative	206.524 ± 0.051	1.0708 ± 0.0502	0.06605 ± 0.00404
Both dependences	Final only	Window 8	206.524 ± 0.051	0.7599 ± 0.0200	0.07684 ± 0.00251
Both dependences	Paired	Reset	233.600 ± 0.000	1.3006 ± 0.0085	0.01346 ± 0.00036
Both dependences	Paired	Cumulative	233.600 ± 0.000	1.4639 ± 0.0232	0.00477 ± 0.00105
Both dependences	Paired	Window 8	233.600 ± 0.000	1.4090 ± 0.0160	0.00786 ± 0.00073
Both dependences	Randomized	Reset	232.500 ± 0.000	1.2746 ± 0.0091	0.02623 ± 0.00115
Both dependences	Randomized	Cumulative	232.500 ± 0.000	1.4414 ± 0.0294	0.00658 ± 0.00135
Both dependences	Randomized	Window 8	232.500 ± 0.000	1.3894 ± 0.0180	0.00951 ± 0.00087
Both dependences	Omit echo	Reset	233.600 ± 0.000	1.1266 ± 0.0064	0.18780 ± 0.00302
Both dependences	Omit echo	Cumulative	233.600 ± 0.000	1.2569 ± 0.0123	0.25611 ± 0.00969
Both dependences	Omit echo	Window 8	233.600 ± 0.000	1.2274 ± 0.0097	0.23109 ± 0.00649
Both dependences	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.07038 ± 0.00000
Both dependences	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.03260 ± 0.00000
Echo shift	Final only	Reset	212.034 ± 0.072	0.4900 ± 0.0051	0.25136 ± 0.00170
Echo shift	Final only	Cumulative	212.034 ± 0.072	1.0534 ± 0.0368	0.11040 ± 0.00621
Echo shift	Final only	Window 8	212.034 ± 0.072	0.8499 ± 0.0191	0.17166 ± 0.00341
Echo shift	Paired	Reset	233.600 ± 0.000	1.4192 ± 0.0018	0.00484 ± 0.00060
Echo shift	Paired	Cumulative	233.600 ± 0.000	1.3526 ± 0.0004	0.00030 ± 0.00018
Echo shift	Paired	Window 8	233.600 ± 0.000	1.3528 ± 0.0004	0.00011 ± 0.00008
Echo shift	Randomized	Reset	232.500 ± 0.000	1.3970 ± 0.0023	0.01227 ± 0.00108
Echo shift	Randomized	Cumulative	232.500 ± 0.000	1.3888 ± 0.0077	0.01105 ± 0.00238
Echo shift	Randomized	Window 8	232.500 ± 0.000	1.3922 ± 0.0045	0.00652 ± 0.00117
Echo shift	Omit echo	Reset	233.600 ± 0.000	1.2530 ± 0.0008	0.12274 ± 0.00071
Echo shift	Omit echo	Cumulative	233.600 ± 0.000	1.2501 ± 0.0001	0.11881 ± 0.00008
Echo shift	Omit echo	Window 8	233.600 ± 0.000	1.2501 ± 0.0001	0.11881 ± 0.00008
Echo shift	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.13724 ± 0.00000
Echo shift	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04849 ± 0.00000
Source asymmetry	Final only	Reset	210.669 ± 0.070	0.4852 ± 0.0054	0.23042 ± 0.00181
Source asymmetry	Final only	Cumulative	210.669 ± 0.070	1.0058 ± 0.0389	0.10114 ± 0.00535
Source asymmetry	Final only	Window 8	210.669 ± 0.070	0.7979 ± 0.0204	0.15384 ± 0.00385
Source asymmetry	Paired	Reset	233.600 ± 0.000	1.4483 ± 0.0014	0.00890 ± 0.00084
Source asymmetry	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0002	0.00024 ± 0.00013
Source asymmetry	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0002	0.00024 ± 0.00013
Source asymmetry	Randomized	Reset	232.500 ± 0.000	1.3701 ± 0.0030	0.02028 ± 0.00107
Source asymmetry	Randomized	Cumulative	232.500 ± 0.000	1.3925 ± 0.0108	0.00640 ± 0.00116
Source asymmetry	Randomized	Window 8	232.500 ± 0.000	1.3752 ± 0.0064	0.00822 ± 0.00069
Source asymmetry	Omit echo	Reset	233.600 ± 0.000	1.2504 ± 0.0006	0.04703 ± 0.00281
Source asymmetry	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.02183 ± 0.00044
Source asymmetry	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.02183 ± 0.00044
Source asymmetry	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.06349 ± 0.00000

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Environment	Policy	Memory	Acquisition cost	Production cost	Regret
Source asymmetry	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04066 ± 0.00000
Stronger humans	Final only	Reset	211.370 ± 0.066	0.5579 ± 0.0063	0.38811 ± 0.00207
Stronger humans	Final only	Cumulative	211.370 ± 0.066	1.4993 ± 0.0418	0.14645 ± 0.00748
Stronger humans	Final only	Window 8	211.370 ± 0.066	1.1133 ± 0.0188	0.24175 ± 0.00576
Stronger humans	Paired	Reset	233.600 ± 0.000	1.5894 ± 0.0041	0.02498 ± 0.00121
Stronger humans	Paired	Cumulative	233.600 ± 0.000	1.5611 ± 0.0008	0.00070 ± 0.00028
Stronger humans	Paired	Window 8	233.600 ± 0.000	1.5614 ± 0.0010	0.00088 ± 0.00032
Stronger humans	Randomized	Reset	232.500 ± 0.000	1.5957 ± 0.0047	0.03157 ± 0.00133
Stronger humans	Randomized	Cumulative	232.500 ± 0.000	1.5609 ± 0.0009	0.00118 ± 0.00041
Stronger humans	Randomized	Window 8	232.500 ± 0.000	1.5618 ± 0.0013	0.00142 ± 0.00043
Stronger humans	Omit echo	Reset	233.600 ± 0.000	1.3492 ± 0.0046	0.17857 ± 0.00304
Stronger humans	Omit echo	Cumulative	233.600 ± 0.000	1.3039 ± 0.0012	0.12595 ± 0.00072
Stronger humans	Omit echo	Window 8	233.600 ± 0.000	1.3053 ± 0.0015	0.12636 ± 0.00082
Stronger humans	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.43790 ± 0.00000
Stronger humans	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.22340 ± 0.00000

C. Final-16-round outcomes

Environment	Policy	Memory	Late error (%)	Late net	Late hit (%)
No echo	Final only	Reset	10.445 ± 0.097	0.62531 ± 0.00303	0.05 ± 0.10
No echo	Final only	Cumulative	3.130 ± 0.228	0.83006 ± 0.00638	27.78 ± 6.96
No echo	Final only	Window 8	7.290 ± 0.265	0.71205 ± 0.00757	5.71 ± 1.04
No echo	Paired	Reset	1.863 ± 0.031	0.85928 ± 0.00096	32.86 ± 2.15
No echo	Paired	Cumulative	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Paired	Window 8	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Randomized	Reset	1.947 ± 0.042	0.85773 ± 0.00129	50.15 ± 2.11
No echo	Randomized	Cumulative	1.770 ± 0.000	0.86349 ± 0.00031	59.33 ± 7.63
No echo	Randomized	Window 8	1.770 ± 0.000	0.86324 ± 0.00021	53.47 ± 5.24
No echo	Omit echo	Reset	1.861 ± 0.031	0.86206 ± 0.00096	98.34 ± 0.57
No echo	Omit echo	Cumulative	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Omit echo	Window 8	1.770 ± 0.000	0.86486 ± 0.00000	100.00 ± 0.00
No echo	Fixed exposed	Fixed	3.520 ± 0.000	0.87440 ± 0.00000	0.00 ± 0.00
No echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
High echo	Final only	Reset	10.631 ± 0.100	0.62109 ± 0.00317	0.00 ± 0.00
High echo	Final only	Cumulative	3.308 ± 0.294	0.82076 ± 0.00821	21.53 ± 6.14
High echo	Final only	Window 8	9.179 ± 0.191	0.65758 ± 0.00552	5.08 ± 1.09
High echo	Paired	Reset	1.895 ± 0.037	0.85708 ± 0.00111	97.61 ± 0.68
High echo	Paired	Cumulative	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
High echo	Paired	Window 8	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
High echo	Randomized	Reset	2.263 ± 0.071	0.84645 ± 0.00212	92.68 ± 1.01
High echo	Randomized	Cumulative	1.770 ± 0.000	0.86113 ± 0.00000	100.00 ± 0.00
High echo	Randomized	Window 8	1.770 ± 0.000	0.86113 ± 0.00000	100.00 ± 0.00
High echo	Omit echo	Reset	9.977 ± 0.049	0.61864 ± 0.00147	0.00 ± 0.00
High echo	Omit echo	Cumulative	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
High echo	Omit echo	Window 8	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
High echo	Fixed exposed	Fixed	9.637 ± 0.000	0.69089 ± 0.00000	0.00 ± 0.00
High echo	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Both dependences	Final only	Reset	10.605 ± 0.090	0.62261 ± 0.00284	0.00 ± 0.00
Both dependences	Final only	Cumulative	9.145 ± 0.183	0.65054 ± 0.00502	8.01 ± 4.00
Both dependences	Final only	Window 8	10.305 ± 0.142	0.62560 ± 0.00439	0.98 ± 0.44
Both dependences	Paired	Reset	7.830 ± 0.030	0.68209 ± 0.00064	51.46 ± 2.24
Both dependences	Paired	Cumulative	7.297 ± 0.047	0.69368 ± 0.00102	92.19 ± 3.82
Both dependences	Paired	Window 8	7.580 ± 0.056	0.68748 ± 0.00119	69.04 ± 4.45
Both dependences	Randomized	Reset	8.269 ± 0.078	0.66957 ± 0.00222	47.17 ± 2.10
Both dependences	Randomized	Cumulative	7.372 ± 0.065	0.69228 ± 0.00141	85.94 ± 5.29

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Environment	Policy	Memory	Late error (%)	Late net	Late hit (%)
Both dependences	Randomized	Window 8	7.605 ± 0.056	0.68720 ± 0.00119	67.09 ± 4.45
Both dependences	Omit echo	Reset	13.735 ± 0.170	0.50836 ± 0.00526	0.00 ± 0.00
Both dependences	Omit echo	Cumulative	16.695 ± 0.321	0.41632 ± 0.00971	0.00 ± 0.00
Both dependences	Omit echo	Window 8	15.002 ± 0.339	0.46839 ± 0.01040	0.00 ± 0.00
Both dependences	Fixed exposed	Fixed	9.919 ± 0.000	0.68242 ± 0.00000	0.00 ± 0.00
Both dependences	Fixed blind	Fixed	8.560 ± 0.000	0.72020 ± 0.00000	0.00 ± 0.00
Echo shift	Final only	Reset	10.644 ± 0.094	0.62065 ± 0.00299	0.00 ± 0.00
Echo shift	Final only	Cumulative	4.285 ± 0.411	0.79352 ± 0.01167	24.41 ± 6.42
Echo shift	Final only	Window 8	9.043 ± 0.201	0.66161 ± 0.00589	6.40 ± 1.25
Echo shift	Paired	Reset	1.890 ± 0.034	0.85722 ± 0.00104	97.80 ± 0.63
Echo shift	Paired	Cumulative	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
Echo shift	Paired	Window 8	1.770 ± 0.000	0.86086 ± 0.00000	100.00 ± 0.00
Echo shift	Randomized	Reset	2.321 ± 0.080	0.84473 ± 0.00238	91.89 ± 1.12
Echo shift	Randomized	Cumulative	1.837 ± 0.057	0.85915 ± 0.00169	99.17 ± 0.71
Echo shift	Randomized	Window 8	1.770 ± 0.000	0.86113 ± 0.00000	100.00 ± 0.00
Echo shift	Omit echo	Reset	9.981 ± 0.048	0.61851 ± 0.00144	0.00 ± 0.00
Echo shift	Omit echo	Cumulative	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
Echo shift	Omit echo	Window 8	9.821 ± 0.000	0.62334 ± 0.00000	0.00 ± 0.00
Echo shift	Fixed exposed	Fixed	9.637 ± 0.000	0.69089 ± 0.00000	0.00 ± 0.00
Echo shift	Fixed blind	Fixed	3.520 ± 0.000	0.87140 ± 0.00000	0.00 ± 0.00
Source asymmetry	Final only	Reset	10.490 ± 0.088	0.62523 ± 0.00279	0.00 ± 0.00
Source asymmetry	Final only	Cumulative	4.391 ± 0.239	0.79002 ± 0.00652	12.65 ± 4.99
Source asymmetry	Final only	Window 8	8.316 ± 0.244	0.68341 ± 0.00695	4.30 ± 1.01
Source asymmetry	Paired	Reset	2.964 ± 0.052	0.82508 ± 0.00158	90.92 ± 1.38
Source asymmetry	Paired	Cumulative	2.685 ± 0.000	0.83342 ± 0.00000	100.00 ± 0.00
Source asymmetry	Paired	Window 8	2.685 ± 0.000	0.83342 ± 0.00000	100.00 ± 0.00
Source asymmetry	Randomized	Reset	3.393 ± 0.054	0.81400 ± 0.00163	49.02 ± 2.20
Source asymmetry	Randomized	Cumulative	2.887 ± 0.055	0.82859 ± 0.00139	75.83 ± 6.61
Source asymmetry	Randomized	Window 8	3.004 ± 0.043	0.82562 ± 0.00107	61.77 ± 5.09
Source asymmetry	Omit echo	Reset	4.310 ± 0.173	0.78864 ± 0.00519	0.00 ± 0.00
Source asymmetry	Omit echo	Cumulative	3.521 ± 0.000	0.81234 ± 0.00000	0.00 ± 0.00
Source asymmetry	Omit echo	Window 8	3.521 ± 0.000	0.81234 ± 0.00000	0.00 ± 0.00
Source asymmetry	Fixed exposed	Fixed	5.101 ± 0.000	0.82697 ± 0.00000	0.00 ± 0.00
Source asymmetry	Fixed blind	Fixed	4.240 ± 0.000	0.84980 ± 0.00000	0.00 ± 0.00
Stronger humans	Final only	Reset	18.684 ± 0.136	0.37746 ± 0.00385	0.00 ± 0.00
Stronger humans	Final only	Cumulative	7.391 ± 0.370	0.69146 ± 0.01067	15.82 ± 5.67
Stronger humans	Final only	Window 8	12.709 ± 0.342	0.54463 ± 0.00987	3.61 ± 1.19
Stronger humans	Paired	Reset	5.481 ± 0.079	0.74670 ± 0.00243	75.88 ± 2.18
Stronger humans	Paired	Cumulative	4.600 ± 0.000	0.77377 ± 0.00000	100.00 ± 0.00
Stronger humans	Paired	Window 8	4.615 ± 0.017	0.77329 ± 0.00052	99.56 ± 0.45
Stronger humans	Randomized	Reset	5.591 ± 0.079	0.74361 ± 0.00240	73.10 ± 2.05
Stronger humans	Randomized	Cumulative	4.600 ± 0.000	0.77404 ± 0.00000	100.00 ± 0.00
Stronger humans	Randomized	Window 8	4.613 ± 0.013	0.77362 ± 0.00041	99.61 ± 0.38
Stronger humans	Omit echo	Reset	10.863 ± 0.191	0.59001 ± 0.00574	0.00 ± 0.00
Stronger humans	Omit echo	Cumulative	8.920 ± 0.000	0.64933 ± 0.00000	0.00 ± 0.00
Stronger humans	Omit echo	Window 8	8.952 ± 0.043	0.64831 ± 0.00129	0.00 ± 0.00
Stronger humans	Fixed exposed	Fixed	19.570 ± 0.000	0.39290 ± 0.00000	0.00 ± 0.00
Stronger humans	Fixed blind	Fixed	12.320 ± 0.000	0.60740 ± 0.00000	0.00 ± 0.00

D. Final-16-round costs and regret

Environment	Policy	Memory	Late acquisition	Late production	Late regret
No echo	Final only	Reset	215.187 ± 0.137	0.4351 ± 0.0094	0.24393 ± 0.00303
No echo	Final only	Cumulative	215.187 ± 0.137	1.1752 ± 0.0603	0.03929 ± 0.00639
No echo	Final only	Window 8	215.187 ± 0.137	0.8326 ± 0.0325	0.15724 ± 0.00757

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Environment	Policy	Memory	Late acquisition	Late production	Late regret
No echo	Paired	Reset	233.600 ± 0.000	1.3864 ± 0.0044	0.00552 ± 0.00096
No echo	Paired	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Paired	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Randomized	Reset	232.500 ± 0.000	1.3511 ± 0.0048	0.00733 ± 0.00129
No echo	Randomized	Cumulative	232.500 ± 0.000	1.3313 ± 0.0153	0.00163 ± 0.00031
No echo	Randomized	Window 8	232.500 ± 0.000	1.3431 ± 0.0105	0.00186 ± 0.00021
No echo	Omit echo	Reset	233.600 ± 0.000	1.2543 ± 0.0015	0.00279 ± 0.00096
No echo	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.00000 ± 0.00000
No echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.04749 ± 0.00000
No echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.05049 ± 0.00000
High echo	Final only	Reset	209.793 ± 0.110	0.4320 ± 0.0089	0.24546 ± 0.00317
High echo	Final only	Cumulative	209.793 ± 0.110	1.4389 ± 0.0728	0.04590 ± 0.00821
High echo	Final only	Window 8	209.793 ± 0.110	0.7887 ± 0.0335	0.20904 ± 0.00553
High echo	Paired	Reset	233.600 ± 0.000	1.4516 ± 0.0013	0.00377 ± 0.00111
High echo	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Randomized	Reset	232.500 ± 0.000	1.4446 ± 0.0025	0.01466 ± 0.00212
High echo	Randomized	Cumulative	232.500 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Randomized	Window 8	232.500 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
High echo	Omit echo	Reset	233.600 ± 0.000	1.2506 ± 0.0008	0.24221 ± 0.00147
High echo	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
High echo	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
High echo	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.22700 ± 0.00000
High echo	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04649 ± 0.00000
Both dependences	Final only	Reset	206.910 ± 0.106	0.4305 ± 0.0082	0.07956 ± 0.00284
Both dependences	Final only	Cumulative	206.910 ± 0.106	1.2295 ± 0.0790	0.05173 ± 0.00502
Both dependences	Final only	Window 8	206.910 ± 0.106	0.7337 ± 0.0334	0.07663 ± 0.00439
Both dependences	Paired	Reset	233.600 ± 0.000	1.2952 ± 0.0152	0.01360 ± 0.00063
Both dependences	Paired	Cumulative	233.600 ± 0.000	1.5183 ± 0.0217	0.00209 ± 0.00102
Both dependences	Paired	Window 8	233.600 ± 0.000	1.4029 ± 0.0269	0.00827 ± 0.00119
Both dependences	Randomized	Reset	232.500 ± 0.000	1.2762 ± 0.0154	0.02638 ± 0.00222
Both dependences	Randomized	Cumulative	232.500 ± 0.000	1.4903 ± 0.0303	0.00376 ± 0.00141
Both dependences	Randomized	Window 8	232.500 ± 0.000	1.3928 ± 0.0275	0.00881 ± 0.00119
Both dependences	Omit echo	Reset	233.600 ± 0.000	1.1245 ± 0.0118	0.18733 ± 0.00527
Both dependences	Omit echo	Cumulative	233.600 ± 0.000	1.2902 ± 0.0084	0.27945 ± 0.00972
Both dependences	Omit echo	Window 8	233.600 ± 0.000	1.2252 ± 0.0164	0.22736 ± 0.01040
Both dependences	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.07038 ± 0.00000
Both dependences	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.03260 ± 0.00000
Echo shift	Final only	Reset	209.670 ± 0.124	0.4359 ± 0.0092	0.24594 ± 0.00299
Echo shift	Final only	Cumulative	209.670 ± 0.124	1.3365 ± 0.0699	0.07317 ± 0.01168
Echo shift	Final only	Window 8	209.670 ± 0.124	0.7926 ± 0.0303	0.20503 ± 0.00589
Echo shift	Paired	Reset	233.600 ± 0.000	1.4524 ± 0.0007	0.00363 ± 0.00104
Echo shift	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Echo shift	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Echo shift	Randomized	Reset	232.500 ± 0.000	1.4427 ± 0.0025	0.01638 ± 0.00238
Echo shift	Randomized	Cumulative	232.500 ± 0.000	1.4483 ± 0.0014	0.00197 ± 0.00169
Echo shift	Randomized	Window 8	232.500 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Echo shift	Omit echo	Reset	233.600 ± 0.000	1.2511 ± 0.0003	0.24234 ± 0.00144
Echo shift	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
Echo shift	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.23752 ± 0.00000
Echo shift	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.22700 ± 0.00000
Echo shift	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04649 ± 0.00000
Source asymmetry	Final only	Reset	211.038 ± 0.117	0.4223 ± 0.0082	0.21360 ± 0.00278
Source asymmetry	Final only	Cumulative	211.038 ± 0.117	1.3368 ± 0.0714	0.04891 ± 0.00653
Source asymmetry	Final only	Window 8	211.038 ± 0.117	0.7773 ± 0.0383	0.15547 ± 0.00696
Source asymmetry	Paired	Reset	233.600 ± 0.000	1.4478 ± 0.0025	0.00832 ± 0.00158

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Environment	Policy	Memory	Late acquisition	Late production	Late regret
Source asymmetry	Paired	Cumulative	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Source asymmetry	Paired	Window 8	233.600 ± 0.000	1.4500 ± 0.0000	0.00000 ± 0.00000
Source asymmetry	Randomized	Reset	232.500 ± 0.000	1.3683 ± 0.0051	0.01962 ± 0.00163
Source asymmetry	Randomized	Cumulative	232.500 ± 0.000	1.4017 ± 0.0132	0.00510 ± 0.00139
Source asymmetry	Randomized	Window 8	232.500 ± 0.000	1.3735 ± 0.0102	0.00806 ± 0.00107
Source asymmetry	Omit echo	Reset	233.600 ± 0.000	1.2506 ± 0.0009	0.04477 ± 0.00519
Source asymmetry	Omit echo	Cumulative	233.600 ± 0.000	1.2500 ± 0.0000	0.02108 ± 0.00000
Source asymmetry	Omit echo	Window 8	233.600 ± 0.000	1.2500 ± 0.0000	0.02108 ± 0.00000
Source asymmetry	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.06349 ± 0.00000
Source asymmetry	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.04066 ± 0.00000
Stronger humans	Final only	Reset	211.641 ± 0.115	0.5113 ± 0.0122	0.40155 ± 0.00386
Stronger humans	Final only	Cumulative	211.641 ± 0.115	1.7568 ± 0.0648	0.08766 ± 0.01067
Stronger humans	Final only	Window 8	211.641 ± 0.115	1.1179 ± 0.0290	0.23444 ± 0.00987
Stronger humans	Paired	Reset	233.600 ± 0.000	1.5895 ± 0.0085	0.02702 ± 0.00243
Stronger humans	Paired	Cumulative	233.600 ± 0.000	1.5600 ± 0.0000	0.00000 ± 0.00000
Stronger humans	Paired	Window 8	233.600 ± 0.000	1.5610 ± 0.0016	0.00048 ± 0.00052
Stronger humans	Randomized	Reset	232.500 ± 0.000	1.5928 ± 0.0076	0.03037 ± 0.00239
Stronger humans	Randomized	Cumulative	232.500 ± 0.000	1.5600 ± 0.0000	0.00000 ± 0.00000
Stronger humans	Randomized	Window 8	232.500 ± 0.000	1.5610 ± 0.0014	0.00042 ± 0.00041
Stronger humans	Omit echo	Reset	233.600 ± 0.000	1.3511 ± 0.0098	0.18371 ± 0.00574
Stronger humans	Omit echo	Cumulative	233.600 ± 0.000	1.3020 ± 0.0000	0.12444 ± 0.00000
Stronger humans	Omit echo	Window 8	233.600 ± 0.000	1.3053 ± 0.0030	0.12545 ± 0.00129
Stronger humans	Fixed exposed	Fixed	0.000 ± 0.000	1.0000 ± 0.0000	0.43790 ± 0.00000
Stronger humans	Fixed blind	Fixed	0.000 ± 0.000	1.1500 ± 0.0000	0.22340 ± 0.00000

E. Paired policy and memory contrasts

Environment	Policy A / memory	Policy B / memory	Error diff. (pp)	Net difference	Regret diff.
No echo	Paired / Cumulative	Omit echo / Cumulative	0.000 ± 0.000	-0.00005 ± 0.00001	0.00005 ± 0.00001
No echo	Paired / Cumulative	Randomized / Cumulative	0.000 ± 0.004	0.00142 ± 0.00027	-0.00168 ± 0.00027
No echo	Paired / Window 8	Paired / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
No echo	Randomized / Window 8	Randomized / Cumulative	0.000 ± 0.000	-0.00012 ± 0.00017	0.00011 ± 0.00017
No echo	Paired / Cumulative	Fixed blind / Fixed	-1.748 ± 0.002	-0.00665 ± 0.00008	-0.05038 ± 0.00008
High echo	Paired / Cumulative	Omit echo / Cumulative	-8.052 ± 0.001	0.23755 ± 0.00004	-0.23755 ± 0.00004
High echo	Paired / Cumulative	Randomized / Cumulative	-0.009 ± 0.008	-0.00000 ± 0.00025	-0.00027 ± 0.00025
High echo	Paired / Window 8	Paired / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
High echo	Randomized / Window 8	Randomized / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
High echo	Paired / Cumulative	Fixed blind / Fixed	-1.747 ± 0.003	-0.01062 ± 0.00009	-0.04641 ± 0.00009
Both dependences	Paired / Cumulative	Omit echo / Cumulative	-8.516 ± 0.365	0.25134 ± 0.01073	-0.25134 ± 0.01073
Both dependences	Paired / Cumulative	Randomized / Cumulative	-0.075 ± 0.061	0.00154 ± 0.00131	-0.00181 ± 0.00131
Both dependences	Paired / Window 8	Paired / Cumulative	0.140 ± 0.034	-0.00310 ± 0.00073	0.00309 ± 0.00073
Both dependences	Randomized / Window 8	Randomized / Cumulative	0.132 ± 0.040	-0.00295 ± 0.00084	0.00293 ± 0.00084
Both dependences	Paired / Cumulative	Fixed blind / Fixed	-1.137 ± 0.049	-0.02921 ± 0.00105	-0.02783 ± 0.00105
Echo shift	Paired / Cumulative	Omit echo / Cumulative	-4.019 ± 0.006	0.11851 ± 0.00017	-0.11851 ± 0.00017
Echo shift	Paired / Cumulative	Randomized / Cumulative	-0.334 ± 0.084	0.01049 ± 0.00239	-0.01075 ± 0.00239
Echo shift	Paired / Window 8	Paired / Cumulative	-0.007 ± 0.006	0.00019 ± 0.00017	-0.00019 ± 0.00017
Echo shift	Randomized / Window 8	Randomized / Cumulative	-0.153 ± 0.076	0.00453 ± 0.00224	-0.00453 ± 0.00224
Echo shift	Paired / Cumulative	Fixed blind / Fixed	-1.741 ± 0.006	-0.00885 ± 0.00018	-0.04819 ± 0.00018
Source asymmetry	Paired / Cumulative	Omit echo / Cumulative	-0.853 ± 0.010	0.02158 ± 0.00031	-0.02158 ± 0.00031
Source asymmetry	Paired / Cumulative	Randomized / Cumulative	-0.244 ± 0.046	0.00590 ± 0.00116	-0.00616 ± 0.00116
Source asymmetry	Paired / Window 8	Paired / Cumulative	0.000 ± 0.000	0.00000 ± 0.00000	0.00000 ± 0.00000
Source asymmetry	Randomized / Window 8	Randomized / Cumulative	0.072 ± 0.030	-0.00183 ± 0.00075	0.00182 ± 0.00074
Source asymmetry	Paired / Cumulative	Fixed blind / Fixed	-1.547 ± 0.004	-0.01662 ± 0.00013	-0.04041 ± 0.00013
Stronger humans	Paired / Cumulative	Omit echo / Cumulative	-4.347 ± 0.016	0.12525 ± 0.00048	-0.12525 ± 0.00048
Stronger humans	Paired / Cumulative	Randomized / Cumulative	-0.016 ± 0.013	0.00021 ± 0.00039	-0.00048 ± 0.00039
Stronger humans	Paired / Window 8	Paired / Cumulative	0.006 ± 0.006	-0.00018 ± 0.00018	0.00018 ± 0.00018
Stronger humans	Randomized / Window 8	Randomized / Cumulative	0.007 ± 0.006	-0.00024 ± 0.00019	0.00024 ± 0.00019
Stronger humans	Paired / Cumulative	Fixed blind / Fixed	-7.697 ± 0.009	0.16567 ± 0.00028	-0.22270 ± 0.00028